# Unit 1: Whole Numbers and Operations

## Exercise 1.1 - Numbers up to 1,000,000 (book page 4)

Try Yourself (book page 2): Write the place and place value of each digit in 237 112, and write it in words.

"2" is at hundred thousands place, so its place value = 2 x 100 000 = 200 000

"3" is at ten thousands place, so its place value = 3 x 10 000 = 30 000

"7" is at thousands place, so its place value = 7 x 1 000 = 7 000

"1" is at hundreds place, so its place value = 1 x 100 = 100

"1" is at tens place, so its place value = 1 x 10 = 10

"2" is at ones place, so its place value = 2 x 1 = 2

Try Yourself (book page 3): K-2 is 861 100 centimetres high. (a) Write it in expanded form. (b) Write the place value of each digit. (c) Write this height in words.

(a) Expanded form: 800 000 + 60 000 + 1 000 + 100.

(b) Place values:

"8" is at hundred thousands place, so its place value = 8 x 100 000 = 800 000

"6" is at ten thousands place, so its place value = 6 x 10 000 = 60 000

"1" is at thousands place, so its place value = 1 x 1 000 = 1 000

"1" is at hundreds place, so its place value = 1 x 100 = 100

"0" is at tens place, so its place value = 0 x 10 = 0

"0" is at ones place, so its place value = 0 x 1 = 0

(c) In words: Eight hundred sixty-one thousand one hundred.

Try Yourself (book page 4): What number am I? I am a 6-digit number. My tens place digit is 5. My ten thousands place digit is 3 less than my tens digit. My ones place digit is the greatest 1-digit even number. My thousands place digit is 3 times my ten thousands place digit. My greatest place value digit is the sum of my tens place digit and my ten thousands place digit. My hundreds place digit is the ones digit of the smallest 2-digit number.

Step 1: Tens digit = 5.

Step 2: Ten thousands digit = 5 - 3 = 2.

Step 3: Ones digit = greatest 1-digit even number = 8.

Step 4: Thousands digit = 3 x 2 = 6.

Step 5: Hundred thousands digit = 5 + 2 = 7.

Step 6: Hundreds digit = ones digit of 10 = 0.

Answer: The number is 726 058.

Q1. Write the following numbers in words:

Write 290 014 in words.

290 014 = Two hundred ninety thousand fourteen.

Write 433 453 in words.

433 453 = Four hundred thirty-three thousand four hundred fifty-three.

Write 910 009 in words.

910 009 = Nine hundred ten thousand nine.

Write 871 653 in words.

871 653 = Eight hundred seventy-one thousand six hundred fifty-three.

Write 242 140 in words.

242 140 = Two hundred forty-two thousand one hundred forty.

Write 688 069 in words.

688 069 = Six hundred eighty-eight thousand sixty-nine.

Write 874 454 in words.

874 454 = Eight hundred seventy-four thousand four hundred fifty-four.

Write 495 523 in words.

495 523 = Four hundred ninety-five thousand five hundred twenty-three.

Q2. Write the following numbers in expanded form:

Write 131 441 in expanded form.

131 441 = 100 000 + 30 000 + 1 000 + 400 + 40 + 1.

Write 600 900 in expanded form.

600 900 = 600 000 + 900.

Write 949 181 in expanded form.

949 181 = 900 000 + 40 000 + 9 000 + 100 + 80 + 1.

Write 466 456 in expanded form.

466 456 = 400 000 + 60 000 + 6 000 + 400 + 50 + 6.

Write 286 019 in expanded form.

286 019 = 200 000 + 80 000 + 6 000 + 10 + 9.

Write 479 321 in expanded form.

479 321 = 400 000 + 70 000 + 9 000 + 300 + 20 + 1.

Write 510 602 in expanded form.

510 602 = 500 000 + 10 000 + 600 + 2.

Write 202 001 in expanded form.

202 001 = 200 000 + 2 000 + 1.

Q3. Match column A with column B.

(a) Eight hundred seven thousand eight hundred = 807 800.

(b) Two hundred seventy-eight thousand seventy-eight = 278 078.

(c) Eight hundred eight thousand eight hundred eight = 808 808.

(d) Two hundred seven thousand five hundred five = 207 505.

(e) Two hundred six thousand two = 206 002.

Q4. Write the place and place value of the coloured digits.

(a) In 545 445, the coloured digit 5 is at hundred thousands place, so its place value = 5 x 100 000 = 500 000.

(b) In 846 532, the coloured digit 3 is at tens place, so its place value = 3 x 10 = 30.

(c) In 682 456, the coloured digit 6 is at hundred thousands place, so its place value = 6 x 100 000 = 600 000.

(d) In 980 714, the coloured digit 8 is at ten thousands place, so its place value = 8 x 10 000 = 80 000.

(e) In 997 924, the coloured digit 9 is at ten thousands place, so its place value = 9 x 10 000 = 90 000.

(f) In 425 419, the coloured digit 4 is at hundreds place, so its place value = 4 x 100 = 400.

(g) In 817 656, the coloured digit 6 is at ones place, so its place value = 6 x 1 = 6.

(h) In 701 232, the coloured digit 3 is at tens place, so its place value = 3 x 10 = 30.

Q5. According to 2017 census, the population of Islamabad is 114 825. (a) Write it in words. (b) Write it in expanded form. (c) Write the place of each digit in it.

(a) In words: One hundred fourteen thousand eight hundred twenty-five.

(b) Expanded form: 100 000 + 10 000 + 4 000 + 800 + 20 + 5.

(c) Places: 1 is at hundred thousands, 1 is at ten thousands, 4 is at thousands, 8 is at hundreds, 2 is at tens, 5 is at ones.

Q6. The speed of light in vacuum is 299 792 kilometres per second. (a) Write it in words. (b) Write it in expanded form. (c) Write the place of each digit in it.

(a) In words: Two hundred ninety-nine thousand seven hundred ninety-two.

(b) Expanded form: 200 000 + 90 000 + 9 000 + 700 + 90 + 2.

(c) Places: 2 is at hundred thousands, 9 is at ten thousands, 9 is at thousands, 7 is at hundreds, 9 is at tens, 2 is at ones.

Exercise 1.2 - Addition and Subtraction (book pages 9-12)

Try Yourself (book page 6): Add 567 098 and 381 940.

8 + 0 = 8

9 + 4 = 13, write 3, carry 1

0 + 9 + 1 (carry) = 10, write 0, carry 1

7 + 1 + 1 (carry) = 9

6 + 8 = 14, write 4, carry 1

5 + 3 + 1 (carry) = 9

Answer: 567 098 + 381 940 = 949 038

Try Yourself (book page 7): There are 124 789 Mathematics books and 200 699 English books in a library. Find the total number of Mathematics and English books.

9 + 9 = 18, write 8, carry 1

8 + 9 + 1 (carry) = 18, write 8, carry 1

7 + 6 + 1 (carry) = 14, write 4, carry 1

4 + 0 + 1 (carry) = 5

2 + 0 = 2

1 + 2 = 3

Answer: 124 789 + 200 699 = 325 488

Try Yourself (book page 7): Add 293 109 and 625 834.

9 + 4 = 13, write 3, carry 1

0 + 3 + 1 (carry) = 4

1 + 8 = 9

3 + 5 = 8

9 + 2 = 11, write 1, carry 1

2 + 6 + 1 (carry) = 9

Answer: 293 109 + 625 834 = 918 943

Try Yourself (book page 7): Find the sum of the smallest 6-digit number and the greatest 6-digit number.

Smallest 6-digit number = 100 000.

Greatest 6-digit number = 999 999.

0 + 9 = 9

0 + 9 = 9

0 + 9 = 9

0 + 9 = 9

0 + 9 = 9

1 + 9 = 10, write 0, carry 1

Carry 1 is written at the front.

Answer: 100 000 + 999 999 = 1 099 999

Try Yourself (book page 8): A toy factory manufactured 598 248 toys, out of which 446 719 toys are of plastic. How many toys are non-plastic toys?

8 is smaller than 9, so borrow 1: 18 - 9 = 9

3 - 1 = 2

2 is smaller than 7, so borrow 1: 12 - 7 = 5

7 - 6 = 1

9 - 4 = 5

5 - 4 = 1

Answer: 598 248 - 446 719 = 151 529

Try Yourself (book page 8): Subtract 289 344 from 760 862.

2 is smaller than 4, so borrow 1: 12 - 4 = 8

5 - 4 = 1

8 - 3 = 5

0 is smaller than 9, so borrow 1: 10 - 9 = 1

6 is smaller than 8, so borrow 1: 15 - 8 = 7

6 - 2 = 4

Answer: 760 862 - 289 344 = 471 518

Try Yourself (book page 8): Find the difference of the smallest 6-digit number and the greatest 5-digit number.

Smallest 6-digit number = 100 000.

Greatest 5-digit number = 99 999.

0 is smaller than 9, so borrow 1: 10 - 9 = 1

0 is smaller than 9, so borrow 1: 9 - 9 = 0

0 is smaller than 9, so borrow 1: 9 - 9 = 0

0 is smaller than 9, so borrow 1: 9 - 9 = 0

0 is smaller than 9, so borrow 1: 9 - 9 = 0

0 - 0 = 0

Answer: 100 000 - 99 999 = 1

Q1. Add the following:

100 700 + 291 562

0 + 2 = 2

0 + 6 = 6

7 + 5 = 12, write 2, carry 1

0 + 1 + 1 (carry) = 2

0 + 9 = 9

1 + 2 = 3

Answer: 100 700 + 291 562 = 392 262

417 381 + 309 201

1 + 1 = 2

8 + 0 = 8

3 + 2 = 5

7 + 9 = 16, write 6, carry 1

1 + 0 + 1 (carry) = 2

4 + 3 = 7

Answer: 417 381 + 309 201 = 726 582

591 727 + 702 929

7 + 9 = 16, write 6, carry 1

2 + 2 + 1 (carry) = 5

7 + 9 = 16, write 6, carry 1

1 + 2 + 1 (carry) = 4

9 + 0 = 9

5 + 7 = 12, write 2, carry 1

Carry 1 is written at the front.

Answer: 591 727 + 702 929 = 1 294 656

319 898 + 428 888

8 + 8 = 16, write 6, carry 1

9 + 8 + 1 (carry) = 18, write 8, carry 1

8 + 8 + 1 (carry) = 17, write 7, carry 1

9 + 8 + 1 (carry) = 18, write 8, carry 1

1 + 2 + 1 (carry) = 4

3 + 4 = 7

Answer: 319 898 + 428 888 = 748 786

766 442 + 611 222

2 + 2 = 4

4 + 2 = 6

4 + 2 = 6

6 + 1 = 7

6 + 1 = 7

7 + 6 = 13, write 3, carry 1

Carry 1 is written at the front.

Answer: 766 442 + 611 222 = 1 377 664

542 001 + 621 416

1 + 6 = 7

0 + 1 = 1

0 + 4 = 4

2 + 1 = 3

4 + 2 = 6

5 + 6 = 11, write 1, carry 1

Carry 1 is written at the front.

Answer: 542 001 + 621 416 = 1 163 417

Q2. Subtract the following:

209 856 - 205 660

6 - 0 = 6

5 is smaller than 6, so borrow 1: 15 - 6 = 9

7 - 6 = 1

9 - 5 = 4

0 - 0 = 0

2 - 2 = 0

Answer: 209 856 - 205 660 = 4 196

788 991 - 206 070

1 - 0 = 1

9 - 7 = 2

9 - 0 = 9

8 - 6 = 2

8 - 0 = 8

7 - 2 = 5

Answer: 788 991 - 206 070 = 582 921

395 108 - 165 439

8 is smaller than 9, so borrow 1: 18 - 9 = 9

0 is smaller than 3, so borrow 1: 9 - 3 = 6

1 is smaller than 4, so borrow 1: 10 - 4 = 6

5 is smaller than 5, so borrow 1: 14 - 5 = 9

8 - 6 = 2

3 - 1 = 2

Answer: 395 108 - 165 439 = 229 669

673 265 - 656 600

5 - 0 = 5

6 - 0 = 6

2 is smaller than 6, so borrow 1: 12 - 6 = 6

3 is smaller than 6, so borrow 1: 12 - 6 = 6

6 - 5 = 1

6 - 6 = 0

Answer: 673 265 - 656 600 = 16 665

686 898 - 333 333

8 - 3 = 5

9 - 3 = 6

8 - 3 = 5

6 - 3 = 3

8 - 3 = 5

6 - 3 = 3

Answer: 686 898 - 333 333 = 353 565

744 762 - 565 656

2 is smaller than 6, so borrow 1: 12 - 6 = 6

5 - 5 = 0

7 - 6 = 1

4 is smaller than 5, so borrow 1: 14 - 5 = 9

4 is smaller than 6, so borrow 1: 13 - 6 = 7

6 - 5 = 1

Answer: 744 762 - 565 656 = 179 106

Q3. Sadia bought a plot for Rs 659 814 and another plot for Rs 799 999. Find the total amount she spent.

4 + 9 = 13, write 3, carry 1

1 + 9 + 1 (carry) = 11, write 1, carry 1

8 + 9 + 1 (carry) = 18, write 8, carry 1

9 + 9 + 1 (carry) = 19, write 9, carry 1

5 + 9 + 1 (carry) = 15, write 5, carry 1

6 + 7 + 1 (carry) = 14, write 4, carry 1

Carry 1 is written at the front.

Answer: 659 814 + 799 999 = 1 459 813

Sadia spent Rs 1 459 813 in total.

Q4. Amaan has an annual income of Rs 456 750. He spends Rs 125 295 on the construction of a Masjid. How much amount is he left with?

0 is smaller than 5, so borrow 1: 10 - 5 = 5

5 is smaller than 9, so borrow 1: 14 - 9 = 5

6 - 2 = 4

6 - 5 = 1

5 - 2 = 3

4 - 1 = 3

Answer: 456 750 - 125 295 = 331 455

Amaan is left with Rs 331 455.

Q5. Lahore has a population of 459 814 in one town and 325 919 in the other. (a) What is the total population of both the towns? (b) What is the difference between their population?

(a) Total population:

4 + 9 = 13, write 3, carry 1

1 + 1 + 1 (carry) = 3

8 + 9 = 17, write 7, carry 1

9 + 5 + 1 (carry) = 15, write 5, carry 1

5 + 2 + 1 (carry) = 8

4 + 3 = 7

Answer: 459 814 + 325 919 = 785 733

(b) Difference:

4 is smaller than 9, so borrow 1: 14 - 9 = 5

1 is smaller than 1, so borrow 1: 10 - 1 = 9

8 is smaller than 9, so borrow 1: 17 - 9 = 8

8 - 5 = 3

5 - 2 = 3

4 - 3 = 1

Answer: 459 814 - 325 919 = 133 895

Q6. The annual yield of a mango orchard is 656 565 kg. In the second year, the yield decreased by 100 984 kg. How much mangoes were produced in the second year?

5 - 4 = 1

6 is smaller than 8, so borrow 1: 16 - 8 = 8

5 is smaller than 9, so borrow 1: 14 - 9 = 5

5 - 0 = 5

5 - 0 = 5

6 - 1 = 5

Answer: 656 565 - 100 984 = 555 581

In the second year, 555 581 kg mangoes were produced.

Q7. The government built 386 655 houses for the homeless in one year. In the second year 24 521 fewer homes were built than the previous year. How many houses did the government build in both the years?

Second year houses:

5 - 1 = 4

5 - 2 = 3

6 - 5 = 1

6 - 4 = 2

8 - 2 = 6

3 - 0 = 3

Answer: 386 655 - 24 521 = 362 134

Total houses in both years:

5 + 4 = 9

5 + 3 = 8

6 + 1 = 7

6 + 2 = 8

8 + 6 = 14, write 4, carry 1

3 + 3 + 1 (carry) = 7

Answer: 386 655 + 362 134 = 748 789

Q8. Fatima has Rs 954 888 in her bank account. If she withdraws Rs 135 600 from the bank to buy a laptop, how much money is she left with?

8 - 0 = 8

8 - 0 = 8

8 - 6 = 2

4 is smaller than 5, so borrow 1: 14 - 5 = 9

4 - 3 = 1

9 - 1 = 8

Answer: 954 888 - 135 600 = 819 288

Fatima is left with Rs 819 288.

Q9. A factory owner gave Rs 448 870 as reward to his employees in one year. In the second year Rs 437 995 were given as reward. (a) What is the total amount given as reward in both the years? (b) In which year did the factory owner give less reward, and how much less?

(a) Total reward:

0 + 5 = 5

7 + 9 = 16, write 6, carry 1

8 + 9 + 1 (carry) = 18, write 8, carry 1

8 + 7 + 1 (carry) = 16, write 6, carry 1

4 + 3 + 1 (carry) = 8

4 + 4 = 8

Answer: 448 870 + 437 995 = 886 865

(b) Less reward was given in the second year:

0 is smaller than 5, so borrow 1: 10 - 5 = 5

7 is smaller than 9, so borrow 1: 16 - 9 = 7

8 is smaller than 9, so borrow 1: 17 - 9 = 8

7 - 7 = 0

4 - 3 = 1

4 - 4 = 0

Answer: 448 870 - 437 995 = 10 875

The second year reward was Rs 10 875 less.

Q10. Farhan donates Rs 600 000 to two Edhi organizations. If he pays Rs 385 990 to one Edhi organization, how much will he pay to the other organization?

0 - 0 = 0

0 is smaller than 9, so borrow 1: 10 - 9 = 1

0 is smaller than 9, so borrow 1: 9 - 9 = 0

0 is smaller than 5, so borrow 1: 9 - 5 = 4

0 is smaller than 8, so borrow 1: 9 - 8 = 1

5 - 3 = 2

Answer: 600 000 - 385 990 = 214 010

Farhan will pay Rs 214 010 to the other organization.

Exercise 1.3 - Multiplication and Division (book pages 13-21)

Try Yourself (book page 11): Multiply 34 523 by 10, 100 and 1 000.

34 523 x 10 = 345 230 (one zero to the right).

34 523 x 100 = 3 452 300 (two zeros to the right).

34 523 x 1 000 = 34 523 000 (three zeros to the right).

Try Yourself (book page 11): Find the product of: (a) 100 x 100 (b) 1000 x 100 x 10 (c) 1000 x 10.

(a) 100 x 100 = 10 000.

(b) 1000 x 100 x 10 = 1 000 000.

(c) 1000 x 10 = 10 000.

Try Yourself (book page 11): The price of a laptop is Rs 102 900. How much do 215 such laptops cost?

102 900 x 5 = 514 500

102 900 x 10 = 1 029 000

102 900 x 200 = 20 580 000

Add the partial products: 514 500 + 1 029 000 + 20 580 000 = 22 123 500

Answer: 102 900 x 215 = 22 123 500

215 laptops cost Rs 22 123 500.

Try Yourself (book page 12): Multiply the greatest 6-digit number by the greatest 3-digit number.

999 999 x 9 = 8 999 991

999 999 x 90 = 89 999 910

999 999 x 900 = 899 999 100

Add the partial products: 8 999 991 + 89 999 910 + 899 999 100 = 998 999 001

Answer: 999 999 x 999 = 998 999 001

Try Yourself (book page 12): Multiply the smallest 6-digit number by the smallest 3-digit number.

100 000 x 100 = 10 000 000.

Try Yourself (book page 12): A company buys 185 motorbikes. If the cost of one motorbike is Rs 79 459, what will be the total cost of 185 such motorbikes?

79 459 x 5 = 397 295

79 459 x 80 = 6 356 720

79 459 x 100 = 7 945 900

Add the partial products: 397 295 + 6 356 720 + 7 945 900 = 14 699 915

Answer: 79 459 x 185 = 14 699 915

185 motorbikes cost Rs 14 699 915.

Try Yourself (book page 12): Find the product of 60 392 and 425.

60 392 x 5 = 301 960

60 392 x 20 = 1 207 840

60 392 x 400 = 24 156 800

Add the partial products: 301 960 + 1 207 840 + 24 156 800 = 25 666 600

Answer: 60 392 x 425 = 25 666 600

Try Yourself (book page 12): Find the product of 23 678 and 32.

23 678 x 2 = 47 356

23 678 x 30 = 710 340

Add the partial products: 47 356 + 710 340 = 757 696

Answer: 23 678 x 32 = 757 696

Try Yourself (book page 13): There are 243 rows of apple trees in a farm. If each row has 21 apple trees, find the total trees in the farm.

243 = 200 + 40 + 3 and 21 = 20 + 1 (box method).

200 x 20 = 4 000, 40 x 20 = 800, 3 x 20 = 60, 200 x 1 = 200, 40 x 1 = 40, 3 x 1 = 3.

4 000 + 800 + 60 + 200 + 40 + 3 = 5 103.

There are 5 103 apple trees in the farm.

Try Yourself (book page 14): Divide 76 000 by 10, 100 and 1 000.

76 000 / 10 = 7 600 (one zero removed from the right).

76 000 / 100 = 760 (two zeros removed).

76 000 / 1 000 = 76 (three zeros removed).

Try Yourself (book page 15): I have saved Rs 16 620 from my pocket money. I want to distribute this amount among 12 children. How much will each child get?

12 x 1 000 = 12 000, remainder 4 620

12 x 300 = 3 600, remainder 1 020

12 x 80 = 960, remainder 60

12 x 5 = 60, remainder 0

Answer: 16 620 / 12 = 1 385

Each child will get Rs 1 385.

Try Yourself (book page 15): Shahzad equally divided Rs 34 760 among 21 needy people. How much amount did each person get? Also find the remaining amount.

21 x 1 000 = 21 000, remainder 13 760

21 x 600 = 12 600, remainder 1 160

21 x 50 = 1 050, remainder 110

21 x 5 = 105, remainder 5

Answer: 34 760 / 21 = 1 655, remainder 5

Each person got Rs 1 655 and Rs 5 remained.

Try Yourself (book page 16): For Pakistan Day celebrations, 10 125 students participated. If groups of 95 students are to be made, find: (a) how many groups can be made? (b) the number of un-grouped students. (c) if 720 students cannot participate, how many groups can be made?

(a) and (b):

95 x 100 = 9 500, remainder 625

95 x 6 = 570, remainder 55

Answer: 10 125 / 95 = 106, remainder 55

Number of groups = 106, un-grouped students = 55.

(c) Remaining students = 10 125 - 720 = 9 405.

95 x 90 = 8 550, remainder 855

95 x 9 = 855, remainder 0

Answer: 9 405 / 95 = 99

So, 99 groups can be made.

Try Yourself (book page 16): If 532 more students participated, then how many groups will be made?

Total students = 10 125 + 532 = 10 657.

95 x 100 = 9 500, remainder 1 157

95 x 10 = 950, remainder 207

95 x 2 = 190, remainder 17

Answer: 10 657 / 95 = 112, remainder 17

112 groups can be made, and 17 students remain un-grouped.

Try Yourself (book page 17): Divide 45 205 by 74. Also find the quotient and remainder.

74 x 600 = 44 400, remainder 805

74 x 10 = 740, remainder 65

Answer: 45 205 / 74 = 610, remainder 65

Q1. Multiply the following numbers by 10, 100 and 1 000:

381 x 10, 100 and 1 000

381 x 10 = 3 810.

381 x 100 = 38 100.

381 x 1 000 = 381 000.

4 090 x 10, 100 and 1 000

4 090 x 10 = 40 900.

4 090 x 100 = 409 000.

4 090 x 1 000 = 4 090 000.

97 509 x 10, 100 and 1 000

97 509 x 10 = 975 090.

97 509 x 100 = 9 750 900.

97 509 x 1 000 = 97 509 000.

69 472 x 10, 100 and 1 000

69 472 x 10 = 694 720.

69 472 x 100 = 6 947 200.

69 472 x 1 000 = 69 472 000.

52 118 x 10, 100 and 1 000

52 118 x 10 = 521 180.

52 118 x 100 = 5 211 800.

52 118 x 1 000 = 52 118 000.

Q2. Divide the following numbers by 10, 100 and 1 000:

49 000 / 10, 100 and 1 000

49 000 / 10 = 4 900.

49 000 / 100 = 490.

49 000 / 1 000 = 49.

78 000 / 10, 100 and 1 000

78 000 / 10 = 7 800.

78 000 / 100 = 780.

78 000 / 1 000 = 78.

65 000 / 10, 100 and 1 000

65 000 / 10 = 6 500.

65 000 / 100 = 650.

65 000 / 1 000 = 65.

97 000 / 10, 100 and 1 000

97 000 / 10 = 9 700.

97 000 / 100 = 970.

97 000 / 1 000 = 97.

21 000 / 10, 100 and 1 000

21 000 / 10 = 2 100.

21 000 / 100 = 210.

21 000 / 1 000 = 21.

Q3. Multiply the following numbers:

624 x 23

624 x 3 = 1 872

624 x 20 = 12 480

Add the partial products: 1 872 + 12 480 = 14 352

Answer: 624 x 23 = 14 352

2 456 x 90

2 456 x 90 = 221 040

Answer: 2 456 x 90 = 221 040

1 092 x 981

1 092 x 1 = 1 092

1 092 x 80 = 87 360

1 092 x 900 = 982 800

Add the partial products: 1 092 + 87 360 + 982 800 = 1 071 252

Answer: 1 092 x 981 = 1 071 252

78 543 x 49

78 543 x 9 = 706 887

78 543 x 40 = 3 141 720

Add the partial products: 706 887 + 3 141 720 = 3 848 607

Answer: 78 543 x 49 = 3 848 607

45 201 x 561

45 201 x 1 = 45 201

45 201 x 60 = 2 712 060

45 201 x 500 = 22 600 500

Add the partial products: 45 201 + 2 712 060 + 22 600 500 = 25 357 761

Answer: 45 201 x 561 = 25 357 761

11 256 x 342

11 256 x 2 = 22 512

11 256 x 40 = 450 240

11 256 x 300 = 3 376 800

Add the partial products: 22 512 + 450 240 + 3 376 800 = 3 849 552

Answer: 11 256 x 342 = 3 849 552

90 902 x 643

90 902 x 3 = 272 706

90 902 x 40 = 3 636 080

90 902 x 600 = 54 541 200

Add the partial products: 272 706 + 3 636 080 + 54 541 200 = 58 449 986

Answer: 90 902 x 643 = 58 449 986

56 219 x 101

56 219 x 1 = 56 219

56 219 x 100 = 5 621 900

Add the partial products: 56 219 + 5 621 900 = 5 678 119

Answer: 56 219 x 101 = 5 678 119

Q4. Divide the following numbers:

13 440 / 15

15 x 800 = 12 000, remainder 1 440

15 x 90 = 1 350, remainder 90

15 x 6 = 90, remainder 0

Answer: 13 440 / 15 = 896

86 449 / 29

29 x 2 000 = 58 000, remainder 28 449

29 x 900 = 26 100, remainder 2 349

29 x 80 = 2 320, remainder 29

29 x 1 = 29, remainder 0

Answer: 86 449 / 29 = 2 981

32 536 / 56

56 x 500 = 28 000, remainder 4 536

56 x 80 = 4 480, remainder 56

56 x 1 = 56, remainder 0

Answer: 32 536 / 56 = 581

47 088 / 48

48 x 900 = 43 200, remainder 3 888

48 x 80 = 3 840, remainder 48

48 x 1 = 48, remainder 0

Answer: 47 088 / 48 = 981

56 780 / 20

20 x 2 000 = 40 000, remainder 16 780
20 x 800 = 16 000, remainder 780
20 x 30 = 600, remainder 180
20 x 9 = 180, remainder 0
Answer: 56 780 / 20 = 2 839

26 166 / 98
98 x 200 = 19 600, remainder 6 566
98 x 60 = 5 880, remainder 686
98 x 7 = 686, remainder 0
Answer: 26 166 / 98 = 267

73 810 / 11
11 x 6 000 = 66 000, remainder 7 810
11 x 700 = 7 700, remainder 110
11 x 10 = 110, remainder 0
Answer: 73 810 / 11 = 6 710

64 454 / 32
32 x 2 000 = 64 000, remainder 454
32 x 10 = 320, remainder 134
32 x 4 = 128, remainder 6
Answer: 64 454 / 32 = 2 014, remainder 6

Q5. Aliya distributed Rs 22 580 among 18 people. How much did each person get, and how much was left?
18 x 1 000 = 18 000, remainder 4 580
18 x 200 = 3 600, remainder 980
18 x 50 = 900, remainder 80
18 x 4 = 72, remainder 8
Answer: 22 580 / 18 = 1 254, remainder 8
Each person got Rs 1 254 and Rs 8 was left.

Q6. Omar's monthly income is Rs 13 582. What is his total income in 134 months?

13 582 x 4 = 54 328

13 582 x 30 = 407 460

13 582 x 100 = 1 358 200

Add the partial products: 54 328 + 407 460 + 1 358 200 = 1 819 988

Answer: 13 582 x 134 = 1 819 988

Omar's total income in 134 months is Rs 1 819 988.

Q7. A toy factory makes 28 550 toys in 25 days. How many toys does it make in one day?

25 x 1 000 = 25 000, remainder 3 550

25 x 100 = 2 500, remainder 1 050

25 x 40 = 1 000, remainder 50

25 x 2 = 50, remainder 0

Answer: 28 550 / 25 = 1 142

The factory makes 1 142 toys in one day.

Q8. A poultry farm produces 76 012 eggs in one day. How many eggs does it produce in 56 days?

76 012 x 6 = 456 072

76 012 x 50 = 3 800 600

Add the partial products: 456 072 + 3 800 600 = 4 256 672

Answer: 76 012 x 56 = 4 256 672

The farm produces 4 256 672 eggs in 56 days.

Q9. The price of one laptop is Rs 89 710. What is the price of 10 such laptops?

89 710 x 10 = 897 100.

10 laptops cost Rs 897 100.

Q10. There are 145 pencil boxes with 5 pencils in each box. 48 boxes are blue. Find the total number of pencils and the number of red pencils.

Total pencils = 145 x 5 = 725.

Red boxes = 145 - 48 = 97.

Red pencils = 97 x 5 = 485.

Total pencils = 725, red pencils = 485.

## Exercise 1.4 - Number Patterns (book pages 22-25)

Try Yourself (book page 20): Ahad planted a plant in a pot. Its height increases by 4 cm daily. If on Monday the height was 12 cm, on which day will the plant be 36 cm high?

Height needed = 36 - 12 = 24 cm more.

Days needed = 24 / 4 = 6 days after Monday.

Monday + 6 days = Sunday.

The plant will be 36 cm high on Sunday.

Try Yourself (book page 21): Identify if this pattern is increasing or decreasing and find the next three terms: 60, 600, 6 000, 60 000, __, __, __.

Rule: multiply by 10 each time (increasing pattern).

60 000 x 10 = 600 000.

600 000 x 10 = 6 000 000.

6 000 000 x 10 = 60 000 000.

Next three terms: 600 000, 6 000 000, 60 000 000.

Try Yourself (book page 23): Identify the rules of these patterns and find the next three terms: (a) 3, 6, 12, 24 (b) 5, 7, 10, 14 (c) 100, 96, 91, 85, 78 (d) 8, 80, 800, 8 000 (e) 900 000, 90 000, 9 000.

(a) Rule: multiply by 2. Next: 48, 96, 192.

(b) Rule: differences increase by 1 each time (+2, +3, +4). Next: 14 + 5 = 19, 19 + 6 = 25, 25 + 7 = 32.

(c) Rule: differences decrease by 1 each time (-4, -5, -6, -7). Next: 78 - 8 = 70, 70 - 9 = 61, 61 - 10 = 51.

(d) Rule: multiply by 10. Next: 80 000, 800 000, 8 000 000.

(e) Rule: divide by 10. Next: 900, 90, 9.

Q1. Identify the rules of these patterns and find the next 3 terms: (a) 10, 40, 160, 640 (b) 22, 220, 2 200 (c) 352, 176, 88 (d) 780, 880, 980 (e) 560, 540, 520, 500.

(a) Rule: multiply by 4. Next: 2 560, 10 240, 40 960.

(b) Rule: multiply by 10. Next: 22 000, 220 000, 2 200 000.

(c) Rule: divide by 2. Next: 44, 22, 11.

(d) Rule: add 100. Next: 1 080, 1 180, 1 280.

(e) Rule: subtract 20. Next: 480, 460, 440.

Q2. Observe the given hundreds chart and identify at least 5 patterns of various arithmetic operations. Also find the rules of these patterns. (Examples: starting at 10 and adding 9 each time gives 10, 19, 28, 37; starting at 100 and subtracting 11 each time gives 100, 89, 78, 67; moving down a column adds 10 each time; moving across a row adds 1 each time; the diagonal starting at 1, 12, 23, 34 adds 11 each time.)

Q3. Observe the given tables and find the rules of the patterns. (The tables show position and term; find how each term is obtained from its position, for example term = position x 40, or term = position x 10, as in Review Exercise Q13.)

## Review Exercise - Unit 1 (book pages 26-28)

Q1. Choose the correct options and fill in the blanks. (a) We put a space after every ___ digits in numbers.

(i) 2

✓ (ii) 3

(iii) 4

(iv) 16

(b) The place value of 2 in the number 985 621 is ___.

(i) 2

✓ (ii) 20

(iii) 200

(iv) 2000

(c) In 856 211, the digit ___ is at thousands place.

(i) 2

(ii) 5

✓ (iii) 6

(iv) 8

(d) When we multiply a number by ___, we put 3 zeros to the right side.

(i) 10

(ii) 100

✓ (iii) 1000

(iv) 1

(e) When we divide a number by ___, we remove one zero from the right side.

✓ (i) 10

(ii) 100

(iii) 1000

(iv) 1

Q2. Write the following numbers in words:

Write 734 123 in words.

734 123 = Seven hundred thirty-four thousand one hundred twenty-three.

Write 965 129 in words.

965 129 = Nine hundred sixty-five thousand one hundred twenty-nine.

Write 982 009 in words.

982 009 = Nine hundred eighty-two thousand nine.

Write 912 011 in words.

912 011 = Nine hundred twelve thousand eleven.

Q3. Solve the following (addition):

212 121 + 56 234

1 + 4 = 5

2 + 3 = 5

1 + 2 = 3

2 + 6 = 8

1 + 5 = 6

2 + 0 = 2

Answer: 212 121 + 56 234 = 268 355

18 315 + 102 376

5 + 6 = 11, write 1, carry 1

1 + 7 + 1 (carry) = 9

3 + 3 = 6

8 + 2 = 10, write 0, carry 1

1 + 0 + 1 (carry) = 2

0 + 1 = 1

Answer: 18 315 + 102 376 = 120 691

727 191 + 92 921

1 + 1 = 2

9 + 2 = 11, write 1, carry 1

1 + 9 + 1 (carry) = 11, write 1, carry 1

7 + 2 + 1 (carry) = 10, write 0, carry 1

2 + 9 + 1 (carry) = 12, write 2, carry 1

7 + 0 + 1 (carry) = 8

Answer: 727 191 + 92 921 = 820 112

139 657 + 247 777

7 + 7 = 14, write 4, carry 1

5 + 7 + 1 (carry) = 13, write 3, carry 1

6 + 7 + 1 (carry) = 14, write 4, carry 1

9 + 7 + 1 (carry) = 17, write 7, carry 1

3 + 4 + 1 (carry) = 8

1 + 2 = 3

Answer: 139 657 + 247 777 = 387 434

532 481 + 100 008

1 + 8 = 9

8 + 0 = 8

4 + 0 = 4

2 + 0 = 2

3 + 0 = 3

5 + 1 = 6

Answer: 532 481 + 100 008 = 632 489

200 454 + 126 654

4 + 4 = 8

5 + 5 = 10, write 0, carry 1

4 + 6 + 1 (carry) = 11, write 1, carry 1

0 + 6 + 1 (carry) = 7

0 + 2 = 2

2 + 1 = 3

Answer: 200 454 + 126 654 = 327 108

Q4. Solve the following (subtraction):

675 921 - 31 412

1 is smaller than 2, so borrow 1: 11 - 2 = 9

1 - 1 = 0

9 - 4 = 5

5 - 1 = 4

7 - 3 = 4

6 - 0 = 6

Answer: 675 921 - 31 412 = 644 509

986 543 - 65 219

3 is smaller than 9, so borrow 1: 13 - 9 = 4

3 - 1 = 2

5 - 2 = 3

6 - 5 = 1

8 - 6 = 2

9 - 0 = 9

Answer: 986 543 - 65 219 = 921 324

108 761 - 70 021

1 - 1 = 0

6 - 2 = 4

7 - 0 = 7

8 - 0 = 8

0 is smaller than 7, so borrow 1: 10 - 7 = 3

0 - 0 = 0

Answer: 108 761 - 70 021 = 38 740

846 109 - 591 089

9 - 9 = 0

0 is smaller than 8, so borrow 1: 10 - 8 = 2

0 - 0 = 0

6 - 1 = 5

4 is smaller than 9, so borrow 1: 14 - 9 = 5

7 - 5 = 2

Answer: 846 109 - 591 089 = 255 020

865 439 - 761 212

9 - 2 = 7

3 - 1 = 2

4 - 2 = 2

5 - 1 = 4

6 - 6 = 0

8 - 7 = 1

Answer: 865 439 - 761 212 = 104 227

696 349 - 288 888

9 - 8 = 1

4 is smaller than 8, so borrow 1: 14 - 8 = 6

3 is smaller than 8, so borrow 1: 12 - 8 = 4

6 is smaller than 8, so borrow 1: 15 - 8 = 7

8 - 8 = 0

6 - 2 = 4

Answer: 696 349 - 288 888 = 407 461

Q5. Solve the following (multiplication):

12 356 x 122

12 356 x 2 = 24 712

12 356 x 20 = 247 120

12 356 x 100 = 1 235 600

Add the partial products: 24 712 + 247 120 + 1 235 600 = 1 507 432

Answer: 12 356 x 122 = 1 507 432

65 781 x 100

65 781 x 100 = 6 578 100

Answer: 65 781 x 100 = 6 578 100

62 825 x 522

62 825 x 2 = 125 650

62 825 x 20 = 1 256 500

62 825 x 500 = 31 412 500

Add the partial products: 125 650 + 1 256 500 + 31 412 500 = 32 794 650

Answer: 62 825 x 522 = 32 794 650

37 564 x 519

37 564 x 9 = 338 076

37 564 x 10 = 375 640

37 564 x 500 = 18 782 000

Add the partial products: 338 076 + 375 640 + 18 782 000 = 19 495 716

Answer: 37 564 x 519 = 19 495 716

Q6. Solve the following (division):

66 693 / 33

33 x 2 000 = 66 000, remainder 693

33 x 20 = 660, remainder 33

33 x 1 = 33, remainder 0

Answer: 66 693 / 33 = 2 021

35 788 / 42

42 x 800 = 33 600, remainder 2 188

42 x 50 = 2 100, remainder 88

42 x 2 = 84, remainder 4

Answer: 35 788 / 42 = 852, remainder 4

25 111 / 69

69 x 300 = 20 700, remainder 4 411

69 x 60 = 4 140, remainder 271

69 x 3 = 207, remainder 64

Answer: 25 111 / 69 = 363, remainder 64

28 000 / 1 000

1 000 x 20 = 20 000, remainder 8 000

1 000 x 8 = 8 000, remainder 0

Answer: 28 000 / 1 000 = 28

58 580 / 10

10 x 5 000 = 50 000, remainder 8 580

10 x 800 = 8 000, remainder 580

10 x 50 = 500, remainder 80

10 x 8 = 80, remainder 0

Answer: 58 580 / 10 = 5 858

28 104 / 28

28 x 1 000 = 28 000, remainder 104

28 x 3 = 84, remainder 20

Answer: 28 104 / 28 = 1 003, remainder 20

Q7. What are the rules for these patterns? Also find the next three terms of each pattern. (a) 50, 100, 150, 200 (b) 180, 165, 150, 135 (c) 18, 90, 450, 2 250 (d) 6 100 000, 610 000, 61 000.

(a) Rule: add 50. Next: 250, 300, 350.

(b) Rule: subtract 15. Next: 120, 105, 90.

(c) Rule: multiply by 5. Next: 11 250, 56 250, 281 250.

(d) Rule: divide by 10. Next: 6 100, 610, 61.

Q8. Find the patterns in the given arithmetic sentences and complete them: 10 x ___ = 10; ___ + 40 = 1 000; ___ + 100 = 100; 10 x ___ = 1 000; 10 x ___ = 10 000.

10 x 1 = 10.

960 + 40 = 1 000.

0 + 100 = 100.

10 x 100 = 1 000.

10 x 1 000 = 10 000.

Q9. The price of a shop is Rs 456 721 and the price of a flat is Rs 987 676. Find the total price of the shop and the flat.

1 + 6 = 7

2 + 7 = 9

7 + 6 = 13, write 3, carry 1

6 + 7 + 1 (carry) = 14, write 4, carry 1

5 + 8 + 1 (carry) = 14, write 4, carry 1

4 + 9 + 1 (carry) = 14, write 4, carry 1

Carry 1 is written at the front.

Answer: 456 721 + 987 676 = 1 444 397

Total price = Rs 1 444 397.

Q10. There are 768 121 children and 456 789 women in the city. How many more children are there than the women?

1 is smaller than 9, so borrow 1: 11 - 9 = 2

2 is smaller than 8, so borrow 1: 11 - 8 = 3

1 is smaller than 7, so borrow 1: 10 - 7 = 3

7 - 6 = 1

6 - 5 = 1

7 - 4 = 3

Answer: 768 121 - 456 789 = 311 332

There are 311 332 more children than women.

Q11. The price of a scanner is Rs 62 900 and the price of a laser printer is Rs 96 880. Find: (a) the total price of both items (b) the total price of 15 scanners and 3 laser printers.

(a) Total price of both:

0 + 0 = 0

0 + 8 = 8

9 + 8 = 17, write 7, carry 1

2 + 6 + 1 (carry) = 9

6 + 9 = 15, write 5, carry 1

Carry 1 is written at the front.

Answer: 62 900 + 96 880 = 159 780

(b) 15 scanners = 62 900 x 15 = 943 500.

3 laser printers = 96 880 x 3 = 290 640.

Total = 943 500 + 290 640 = 1 234 140.

Total price of 15 scanners and 3 laser printers = Rs 1 234 140.

Q12. 35 288 blocks are to be packed in 28 boxes. Find: (a) how many blocks are there in each box? (b) how many blocks will be left?

28 x 1 000 = 28 000, remainder 7 288

28 x 200 = 5 600, remainder 1 688

28 x 60 = 1 680, remainder 8

Answer: 35 288 / 28 = 1 260, remainder 8

(a) Each box has 1 260 blocks. (b) 8 blocks are left.

Q13. Observe the given tables and find the rules of the patterns given in them.

First table: Position 1, 2, 3, 4 gives Term 40, 80, 120, 160. Rule: Term = Position x 40.

Second table: Position 1, 2, 3, 4 gives Term 10, 20, 30, 40. Rule: Term = Position x 10.

## Unit 2: HCF and LCM

### Exercise 2.1 - Highest Common Factor (book pages 30-33)

Try Yourself (book page 29): Nida wants to plant 12 rose plants and 18 jasmine plants in rows in her home garden. If she wants to plant the same type of plant in one row, find the maximum number of plants that can be grown in one row.

Prime factorization of 12 = 2 x 2 x 3.

Prime factorization of 18 = 2 x 3 x 3.

Common prime factors: 2 x 3 = 6.

Answer: HCF = 6.

Maximum plants in one row = 6.

Worked example (book page 30): Sara has two pieces of ribbon whose lengths are 18 cm and 24 cm. She wants to cut the ribbon into smaller pieces of equal length without any ribbon left. What will be the greatest possible length of each piece?

Prime factorization of 18 = 2 x 3 x 3.

Prime factorization of 24 = 2 x 2 x 2 x 3.

Common prime factors: 2 x 3 = 6.

Answer: HCF = 6.

Greatest possible length of each piece = 6 cm.

Worked example (book page 31): Find the HCF of 12, 30 and 44 by using prime factorization.

Prime factorization of 12 = 2 x 2 x 3.

Prime factorization of 30 = 2 x 3 x 5.

Prime factorization of 44 = 2 x 2 x 11.

Common prime factors: 2 = 2.

Answer: HCF = 2.

Worked example (book page 31): Hammad has 36 red pencils and 54 blue pencils. He wants to put these pencils in boxes such that every box has an equal number of pencils of the same colour. What will be the maximum number of pencils in each box?

Prime factorization of 36 = 2 x 2 x 3 x 3.

Prime factorization of 54 = 2 x 3 x 3 x 3.

Division method steps:

54 = 36 x 1 + 18

36 = 18 x 2 + 0

The last divisor is 18, so the HCF = 18.

Answer: HCF = 18.

Maximum pencils in each box = 18.

Worked example (book page 32): Find the greatest number which completely divides 26, 48 and 60.

Prime factorization of 26 = 2 x 13.

Prime factorization of 48 = 2 x 2 x 2 x 2 x 3.

Prime factorization of 60 = 2 x 2 x 3 x 5.

Division method steps:

60 = 26 x 2 + 8

26 = 8 x 3 + 2

8 = 2 x 4 + 0

60 = 2 x 30 + 0

The last divisor is 2, so the HCF = 2.

Answer: HCF = 2.

Try It (book page 32): Find three 2-digit numbers whose sum is 152 and whose HCF is 8.

Take 24, 56 and 72. Their sum = 24 + 56 + 72 = 152.

Prime factorization of 24 = 2 x 2 x 2 x 3.

Prime factorization of 56 = 2 x 2 x 2 x 7.

Prime factorization of 72 = 2 x 2 x 2 x 3 x 3.

Common prime factors: 2 x 2 x 2 = 8.

Answer: HCF = 8.

So 24, 56 and 72 are three such numbers.

Q1. Find the HCF of the following numbers by using prime factorization method:

HCF of 58, 72

Prime factorization of 58 = 2 x 29.

Prime factorization of 72 = 2 x 2 x 2 x 3 x 3.

Common prime factors: 2 = 2.

Answer: HCF = 2.

HCF of 21, 48

Prime factorization of 21 = 3 x 7.

Prime factorization of 48 = 2 x 2 x 2 x 2 x 3.

Common prime factors: 3 = 3.

Answer: HCF = 3.

HCF of 56, 70

Prime factorization of 56 = 2 x 2 x 2 x 7.

Prime factorization of 70 = 2 x 5 x 7.

Common prime factors: 2 x 7 = 14.

Answer: HCF = 14.

HCF of 45, 90

Prime factorization of 45 = 3 x 3 x 5.

Prime factorization of 90 = 2 x 3 x 3 x 5.

Common prime factors: 3 x 3 x 5 = 45.

Answer: HCF = 45.

HCF of 42, 49

Prime factorization of 42 = 2 x 3 x 7.

Prime factorization of 49 = 7 x 7.

Common prime factors: 7 = 7.

Answer: HCF = 7.

HCF of 15, 18, 56

Prime factorization of 15 = 3 x 5.

Prime factorization of 18 = 2 x 3 x 3.

Prime factorization of 56 = 2 x 2 x 2 x 7.

There is no common prime factor, so the HCF = 1.

Answer: HCF = 1.

HCF of 42, 54, 64

Prime factorization of 42 = 2 x 3 x 7.

Prime factorization of 54 = 2 x 3 x 3 x 3.

Prime factorization of 64 = 2 x 2 x 2 x 2 x 2 x 2.

Common prime factors: 2 = 2.

Answer: HCF = 2.

HCF of 18, 30, 90

Prime factorization of 18 = 2 x 3 x 3.

Prime factorization of 30 = 2 x 3 x 5.

Prime factorization of 90 = 2 x 3 x 3 x 5.

Common prime factors: 2 x 3 = 6.

Answer: HCF = 6.

HCF of 12, 24, 36

Prime factorization of 12 = 2 x 2 x 3.

Prime factorization of 24 = 2 x 2 x 2 x 3.

Prime factorization of 36 = 2 x 2 x 3 x 3.

Common prime factors: 2 x 2 x 3 = 12.

Answer: HCF = 12.

HCF of 18, 36, 76

Prime factorization of 18 = 2 x 3 x 3.

Prime factorization of 36 = 2 x 2 x 3 x 3.

Prime factorization of 76 = 2 x 2 x 19.

Common prime factors: 2 = 2.

Answer: HCF = 2.

HCF of 5, 35, 40

Prime factorization of 5 = 5.

Prime factorization of 35 = 5 x 7.

Prime factorization of 40 = 2 x 2 x 2 x 5.

Common prime factors: 5 = 5.

Answer: HCF = 5.

HCF of 13, 52, 78

Prime factorization of 13 = 13.

Prime factorization of 52 = 2 x 2 x 13.

Prime factorization of 78 = 2 x 3 x 13.

Common prime factors: 13 = 13.

Answer: HCF = 13.

Q2. Find the HCF of the following numbers by using division method:

HCF of 13, 65

Prime factorization of 13 = 13.

Prime factorization of 65 = 5 x 13.

Division method steps:

65 = 13 x 5 + 0

The last divisor is 13, so the HCF = 13.

Answer: HCF = 13.

HCF of 25, 75

Prime factorization of 25 = 5 x 5.

Prime factorization of 75 = 3 x 5 x 5.

Division method steps:

75 = 25 x 3 + 0

The last divisor is 25, so the HCF = 25.

Answer: HCF = 25.

HCF of 42, 98

Prime factorization of 42 = 2 x 3 x 7.

Prime factorization of 98 = 2 x 7 x 7.

Division method steps:

98 = 42 x 2 + 14

42 = 14 x 3 + 0

The last divisor is 14, so the HCF = 14.

Answer: HCF = 14.

HCF of 16, 20, 70

Prime factorization of 16 = 2 x 2 x 2 x 2.

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 70 = 2 x 5 x 7.

Division method steps:

70 = 16 x 4 + 6

16 = 6 x 2 + 4

6 = 4 x 1 + 2

4 = 2 x 2 + 0

70 = 4 x 17 + 2

4 = 2 x 2 + 0

The last divisor is 2, so the HCF = 2.

Answer: HCF = 2.

HCF of 56, 84, 88

Prime factorization of 56 = 2 x 2 x 2 x 7.

Prime factorization of 84 = 2 x 2 x 3 x 7.

Prime factorization of 88 = 2 x 2 x 2 x 11.

Division method steps:

88 = 56 x 1 + 32

56 = 32 x 1 + 24

32 = 24 x 1 + 8

24 = 8 x 3 + 0

88 = 28 x 3 + 4

28 = 4 x 7 + 0

The last divisor is 4, so the HCF = 4.

Answer: HCF = 4.

HCF of 57, 76, 95

Prime factorization of 57 = 3 x 19.

Prime factorization of 76 = 2 x 2 x 19.

Prime factorization of 95 = 5 x 19.

Division method steps:

95 = 57 x 1 + 38

57 = 38 x 1 + 19

38 = 19 x 2 + 0

95 = 19 x 5 + 0

The last divisor is 19, so the HCF = 19.

Answer: HCF = 19.

HCF of 16, 32, 96

Prime factorization of 16 = 2 x 2 x 2 x 2.

Prime factorization of 32 = 2 x 2 x 2 x 2 x 2.

Prime factorization of 96 = 2 x 2 x 2 x 2 x 2 x 3.

Division method steps:

96 = 16 x 6 + 0

96 = 16 x 6 + 0

The last divisor is 16, so the HCF = 16.

Answer: HCF = 16.

HCF of 20, 40, 80

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 40 = 2 x 2 x 2 x 5.

Prime factorization of 80 = 2 x 2 x 2 x 2 x 5.

Division method steps:

80 = 20 x 4 + 0

80 = 20 x 4 + 0

The last divisor is 20, so the HCF = 20.

Answer: HCF = 20.

HCF of 48, 76, 96

Prime factorization of 48 = 2 x 2 x 2 x 2 x 3.

Prime factorization of 76 = 2 x 2 x 19.

Prime factorization of 96 = 2 x 2 x 2 x 2 x 2 x 3.

Division method steps:

96 = 48 x 2 + 0

96 = 4 x 24 + 0

The last divisor is 4, so the HCF = 4.

Answer: HCF = 4.

HCF of 24, 48, 72

Prime factorization of 24 = 2 x 2 x 2 x 3.

Prime factorization of 48 = 2 x 2 x 2 x 2 x 3.

Prime factorization of 72 = 2 x 2 x 2 x 3 x 3.

Division method steps:

72 = 24 x 3 + 0

72 = 24 x 3 + 0

The last divisor is 24, so the HCF = 24.

Answer: HCF = 24.

HCF of 51, 65, 75

Prime factorization of 51 = 3 x 17.

Prime factorization of 65 = 5 x 13.

Prime factorization of 75 = 3 x 5 x 5.

Division method steps:

75 = 51 x 1 + 24

51 = 24 x 2 + 3

24 = 3 x 8 + 0

75 = 1 x 75 + 0

The last divisor is 1, so the HCF = 1.

Answer: HCF = 1.

HCF of 13, 39, 78

Prime factorization of 13 = 13.

Prime factorization of 39 = 3 x 13.

Prime factorization of 78 = 2 x 3 x 13.

Division method steps:

78 = 13 x 6 + 0

78 = 13 x 6 + 0

The last divisor is 13, so the HCF = 13.

Answer: HCF = 13.

Q3. The lengths of the two ropes are 24 metres and 14 metres respectively. If Ali wants to cut the ropes into pieces of equal length without any rope left, find out what will be the maximum length of each piece?

Prime factorization of 24 = 2 x 2 x 2 x 3.

Prime factorization of 14 = 2 x 7.

Common prime factors: 2 = 2.

Answer: HCF = 2.

Maximum length of each piece = 2 metres.

Q4. For the Independence Day celebrations, 52 students in white dress, 65 students in green dress and 39 students in blue dress are to be arranged in equal rows in such a way that students of one colour dress will be in each row. What is the greatest number of students that could be in each row?

Prime factorization of 52 = 2 x 2 x 13.

Prime factorization of 65 = 5 x 13.

Prime factorization of 39 = 3 x 13.

Common prime factors: 13 = 13.

Answer: HCF = 13.

Greatest number of students in each row = 13.

Q5. Find the greatest number that divides 16, 24 and 48 completely.

Prime factorization of 16 = 2 x 2 x 2 x 2.

Prime factorization of 24 = 2 x 2 x 2 x 3.

Prime factorization of 48 = 2 x 2 x 2 x 2 x 3.

Common prime factors: 2 x 2 x 2 = 8.

Answer: HCF = 8.

Q6. Ibrahim and Mehwish are preparing first aid kits for the students. They have 30 perforated adhesive bandages, 60 triangular bandages and 75 rectangular bandages. They must distribute these equally in the kits, with nothing left over. What is the greatest number of kits they can be made with this quantity of bandages?

Prime factorization of 30 = 2 x 3 x 5.

Prime factorization of 60 = 2 x 2 x 3 x 5.

Prime factorization of 75 = 3 x 5 x 5.

Common prime factors: 3 x 5 = 15.

Answer: HCF = 15.

Greatest number of kits = 15.

## Exercise 2.2 - Least Common Multiple (book pages 34-37)

Worked example (book page 33): Grade 4 students visit the science laboratory after every 8 days. Grade 5 students visit after every 12 days. If the students of both grades visited the laboratory today, when will they again visit the laboratory on the same day?

Multiples of 8: 8, 16, 24, 32, 40, 48.

Multiples of 12: 12, 24, 36, 48.

First common multiple = 24.

Prime factorization of 8 = 2 x 2 x 2.

Prime factorization of 12 = 2 x 2 x 3.

Answer: LCM = 24.

They will visit together again on the 24th day.

Worked example (book page 35): Find the LCM of 16, 30 and 64 by using prime factorization.

Prime factorization of 16 = 2 x 2 x 2 x 2.

Prime factorization of 30 = 2 x 3 x 5.

Prime factorization of 64 = 2 x 2 x 2 x 2 x 2 x 2.

Answer: LCM = 960.

Worked example (book page 35): Ali, Ahmad and Umar exercise after every 10, 18 and 20 days respectively. If they all are exercising today, when will they exercise on the same day again?

Prime factorization of 10 = 2 x 5.

Prime factorization of 18 = 2 x 3 x 3.

Prime factorization of 20 = 2 x 2 x 5.

Answer: LCM = 180.

They will exercise together again on the 180th day.

Worked example (book page 36): Find the LCM of 12, 20 and 30 by using division.

Prime factorization of 12 = 2 x 2 x 3.

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 30 = 2 x 3 x 5.

Answer: LCM = 60.

Try It (book page 36): A welfare organization is distributing bundles of basic safety items. A packet of masks has 25 masks, a packet of gloves has 20 pairs of gloves and a packet of sanitizer has 5 sanitizer bottles. Find the minimum number of packets of each item so that every bundle has one mask, one glove pair and one sanitizer bottle in it and no object is left over.

Prime factorization of 25 = 5 x 5.

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 5 = 5.

Answer: LCM = 100.

Packets needed: masks = 100 / 25 = 4 packets, gloves = 100 / 20 = 5 packets, sanitizer = 100 / 5 = 20 packets.

Q1. Find the LCM of the following numbers using prime factorization:

LCM of 3, 21

Prime factorization of 3 = 3.

Prime factorization of 21 = 3 x 7.

Answer: LCM = 21.

LCM of 12, 80

Prime factorization of 12 = 2 x 2 x 3.

Prime factorization of 80 = 2 x 2 x 2 x 2 x 5.

Answer: LCM = 240.

LCM of 20, 15

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 15 = 3 x 5.

Answer: LCM = 60.

LCM of 4, 10, 16

Prime factorization of 4 = 2 x 2.

Prime factorization of 10 = 2 x 5.

Prime factorization of 16 = 2 x 2 x 2 x 2.

Answer: LCM = 80.

LCM of 9, 18, 27

Prime factorization of 9 = 3 x 3.

Prime factorization of 18 = 2 x 3 x 3.

Prime factorization of 27 = 3 x 3 x 3.

Answer: LCM = 54.

LCM of 10, 20, 35

Prime factorization of 10 = 2 x 5.

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 35 = 5 x 7.

Answer: LCM = 140.

LCM of 20, 60, 75

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 60 = 2 x 2 x 3 x 5.

Prime factorization of 75 = 3 x 5 x 5.

Answer: LCM = 300.

LCM of 30, 45, 90

Prime factorization of 30 = 2 x 3 x 5.

Prime factorization of 45 = 3 x 3 x 5.

Prime factorization of 90 = 2 x 3 x 3 x 5.

Answer: LCM = 90.

LCM of 16, 24, 36

Prime factorization of 16 = 2 x 2 x 2 x 2.

Prime factorization of 24 = 2 x 2 x 2 x 3.

Prime factorization of 36 = 2 x 2 x 3 x 3.

Answer: LCM = 144.

LCM of 18, 60, 75

Prime factorization of 18 = 2 x 3 x 3.

Prime factorization of 60 = 2 x 2 x 3 x 5.

Prime factorization of 75 = 3 x 5 x 5.

Answer: LCM = 900.

LCM of 49, 51, 56

Prime factorization of 49 = 7 x 7.

Prime factorization of 51 = 3 x 17.

Prime factorization of 56 = 2 x 2 x 2 x 7.

Answer: LCM = 19 992.

LCM of 13, 65, 71

Prime factorization of 13 = 13.

Prime factorization of 65 = 5 x 13.

Prime factorization of 71 = 71.

Answer: LCM = 4 615.

Q2. Find the LCM of the following numbers using division:

LCM of 14, 70

Prime factorization of 14 = 2 x 7.

Prime factorization of 70 = 2 x 5 x 7.

Answer: LCM = 70.

LCM of 15, 30

Prime factorization of 15 = 3 x 5.

Prime factorization of 30 = 2 x 3 x 5.

Answer: LCM = 30.

LCM of 45, 90

Prime factorization of 45 = 3 x 3 x 5.

Prime factorization of 90 = 2 x 3 x 3 x 5.

Answer: LCM = 90.

LCM of 35, 60, 75

Prime factorization of 35 = 5 x 7.

Prime factorization of 60 = 2 x 2 x 3 x 5.

Prime factorization of 75 = 3 x 5 x 5.

Answer: LCM = 2 100.

LCM of 7, 21, 49

Prime factorization of 7 = 7.

Prime factorization of 21 = 3 x 7.

Prime factorization of 49 = 7 x 7.

Answer: LCM = 147.

LCM of 25, 45, 95

Prime factorization of 25 = 5 x 5.

Prime factorization of 45 = 3 x 3 x 5.

Prime factorization of 95 = 5 x 19.

Answer: LCM = 4 275.

LCM of 16, 32, 48

Prime factorization of 16 = 2 x 2 x 2 x 2.

Prime factorization of 32 = 2 x 2 x 2 x 2 x 2.

Prime factorization of 48 = 2 x 2 x 2 x 2 x 3.

Answer: LCM = 96.

LCM of 28, 32, 40

Prime factorization of 28 = 2 x 2 x 7.

Prime factorization of 32 = 2 x 2 x 2 x 2 x 2.

Prime factorization of 40 = 2 x 2 x 2 x 5.

Answer: LCM = 1 120.

LCM of 12, 14, 26

Prime factorization of 12 = 2 x 2 x 3.

Prime factorization of 14 = 2 x 7.

Prime factorization of 26 = 2 x 13.

Answer: LCM = 1 092.

LCM of 10, 20, 25

Prime factorization of 10 = 2 x 5.

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 25 = 5 x 5.

Answer: LCM = 100.

LCM of 7, 14, 21

Prime factorization of 7 = 7.

Prime factorization of 14 = 2 x 7.

Prime factorization of 21 = 3 x 7.

Answer: LCM = 42.

LCM of 8, 32, 42

Prime factorization of 8 = 2 x 2 x 2.

Prime factorization of 32 = 2 x 2 x 2 x 2 x 2.

Prime factorization of 42 = 2 x 3 x 7.

Answer: LCM = 672.

Q3. Find the minimum length of the ribbon which can completely be cut into pieces of length 45 cm, 75 cm and 85 cm without any left over.

Prime factorization of 45 = 3 x 3 x 5.

Prime factorization of 75 = 3 x 5 x 5.

Prime factorization of 85 = 5 x 17.

Answer: LCM = 3 825.

Minimum length of the ribbon = 3 825 cm.

Q4. Find the smallest number that can completely be divided by 42, 38 and 16 without leaving any remainder.

Prime factorization of 42 = 2 x 3 x 7.

Prime factorization of 38 = 2 x 19.

Prime factorization of 16 = 2 x 2 x 2 x 2.

Answer: LCM = 6 384.

Q5. The tour buses for Badshahi Masjid leave the station every 25 minutes, for the interior city every 15 minutes and for the zoo every 30 minutes. If the three buses leave the station simultaneously at 11:05 a.m., find out the time when the three buses will next leave the station simultaneously?

Prime factorization of 25 = 5 x 5.

Prime factorization of 15 = 3 x 5.

Prime factorization of 30 = 2 x 3 x 5.

Answer: LCM = 150.

150 minutes = 2 hours 30 minutes.

11:05 a.m. + 2 hours 30 minutes = 1:35 p.m.

The three buses will next leave together at 1:35 p.m.

Q6. Boxes having 22 centimetre, 35 centimetre and 50 centimetre height respectively are to be stacked next to each other. What is the shortest possible height at which the three types of boxes will be at the same height?

Prime factorization of 22 = 2 x 11.

Prime factorization of 35 = 5 x 7.

Prime factorization of 50 = 2 x 5 x 5.

Answer: LCM = 3 850.

Shortest possible height = 3 850 cm.

Q7. Students of grade 5 have Mathematics test after every 3 days, English test after every 6 days and Science test after every 9 days. If all the three tests were conducted today, find out when will the three tests be conducted next on the same day?

Prime factorization of 3 = 3.

Prime factorization of 6 = 2 x 3.

Prime factorization of 9 = 3 x 3.

Answer: LCM = 18.

The three tests will next fall on the same day after 18 days.

## Review Exercise - Unit 2 (book pages 38-39)

Q1. (a) The HCF of 20, 48 and 56 is ___.

✓ (i) 4

(ii) 3

(iii) 5

(iv) 1

(b) The HCF of two or more numbers, which have no common prime factor, is always ___.

(i) 1000

(ii) 100

(iii) 10

✓ (iv) 1

(c) Prime factorization of 16 is ___.

(i) 2 x 8

(ii) 1 x 16

✓ (iii) 2 x 2 x 2 x 2

(iv) 2 x 4 x 2

(d) The LCM of 33, 66 and 81 is ___.

(i) 1770

(ii) 1872

✓ (iii) 1782

(iv) 1287

(e) The LCM of two or more prime numbers is always equal to their ___.

(i) Prime factor

(ii) Quotient

(iii) LCM

✓ (iv) Product

Q2. Find the HCF of the following numbers by using prime factorization:

HCF of 15, 18

Prime factorization of 15 = 3 x 5.

Prime factorization of 18 = 2 x 3 x 3.

Common prime factors: 3 = 3.

Answer: HCF = 3.

HCF of 10, 20

Prime factorization of 10 = 2 x 5.

Prime factorization of 20 = 2 x 2 x 5.

Common prime factors: 2 x 5 = 10.

Answer: HCF = 10.

HCF of 25, 40

Prime factorization of 25 = 5 x 5.

Prime factorization of 40 = 2 x 2 x 2 x 5.

Common prime factors: 5 = 5.

Answer: HCF = 5.

HCF of 56, 88

Prime factorization of 56 = 2 x 2 x 2 x 7.

Prime factorization of 88 = 2 x 2 x 2 x 11.

Common prime factors: 2 x 2 x 2 = 8.

Answer: HCF = 8.

HCF of 10, 18, 22

Prime factorization of 10 = 2 x 5.

Prime factorization of 18 = 2 x 3 x 3.

Prime factorization of 22 = 2 x 11.

Common prime factors: 2 = 2.

Answer: HCF = 2.

HCF of 20, 40, 82

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 40 = 2 x 2 x 2 x 5.

Prime factorization of 82 = 2 x 41.

Common prime factors: 2 = 2.

Answer: HCF = 2.

HCF of 16, 38, 98

Prime factorization of 16 = 2 x 2 x 2 x 2.

Prime factorization of 38 = 2 x 19.

Prime factorization of 98 = 2 x 7 x 7.

Common prime factors: 2 = 2.

Answer: HCF = 2.

HCF of 39, 51, 75

Prime factorization of 39 = 3 x 13.

Prime factorization of 51 = 3 x 17.

Prime factorization of 75 = 3 x 5 x 5.

Common prime factors: 3 = 3.

Answer: HCF = 3.

Q3. Find the HCF of the following numbers using division:

HCF of 20, 50

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 50 = 2 x 5 x 5.

Division method steps:

50 = 20 x 2 + 10

20 = 10 x 2 + 0

The last divisor is 10, so the HCF = 10.

Answer: HCF = 10.

HCF of 15, 45

Prime factorization of 15 = 3 x 5.

Prime factorization of 45 = 3 x 3 x 5.

Division method steps:

45 = 15 x 3 + 0

The last divisor is 15, so the HCF = 15.

Answer: HCF = 15.

HCF of 60, 70, 80

Prime factorization of 60 = 2 x 2 x 3 x 5.

Prime factorization of 70 = 2 x 5 x 7.

Prime factorization of 80 = 2 x 2 x 2 x 2 x 5.

Division method steps:

80 = 60 x 1 + 20

60 = 20 x 3 + 0

80 = 10 x 8 + 0

The last divisor is 10, so the HCF = 10.

Answer: HCF = 10.

HCF of 22, 28, 32

Prime factorization of 22 = 2 x 11.

Prime factorization of 28 = 2 x 2 x 7.

Prime factorization of 32 = 2 x 2 x 2 x 2 x 2.

Division method steps:

32 = 22 x 1 + 10

22 = 10 x 2 + 2

10 = 2 x 5 + 0

32 = 2 x 16 + 0

The last divisor is 2, so the HCF = 2.

Answer: HCF = 2.

HCF of 44, 55, 99

Prime factorization of 44 = 2 x 2 x 11.

Prime factorization of 55 = 5 x 11.

Prime factorization of 99 = 3 x 3 x 11.

Division method steps:

99 = 44 x 2 + 11

44 = 11 x 4 + 0

99 = 11 x 9 + 0

The last divisor is 11, so the HCF = 11.

Answer: HCF = 11.

HCF of 34, 48, 62

Prime factorization of 34 = 2 x 17.

Prime factorization of 48 = 2 x 2 x 2 x 2 x 3.

Prime factorization of 62 = 2 x 31.

Division method steps:

62 = 34 x 1 + 28

34 = 28 x 1 + 6

28 = 6 x 4 + 4

6 = 4 x 1 + 2

4 = 2 x 2 + 0

62 = 2 x 31 + 0

The last divisor is 2, so the HCF = 2.

Answer: HCF = 2.

HCF of 30, 45, 70

Prime factorization of 30 = 2 x 3 x 5.

Prime factorization of 45 = 3 x 3 x 5.

Prime factorization of 70 = 2 x 5 x 7.

Division method steps:

70 = 30 x 2 + 10

30 = 10 x 3 + 0

70 = 15 x 4 + 10

15 = 10 x 1 + 5

10 = 5 x 2 + 0

The last divisor is 5, so the HCF = 5.

Answer: HCF = 5.

HCF of 26, 52, 65

Prime factorization of 26 = 2 x 13.

Prime factorization of 52 = 2 x 2 x 13.

Prime factorization of 65 = 5 x 13.

Division method steps:

65 = 26 x 2 + 13

26 = 13 x 2 + 0

65 = 26 x 2 + 13

26 = 13 x 2 + 0

The last divisor is 13, so the HCF = 13.

Answer: HCF = 13.

Q4. Find the LCM of the following numbers using prime factorization:

LCM of 2, 5

Prime factorization of 2 = 2.

Prime factorization of 5 = 5.

Answer: LCM = 10.

LCM of 3, 7

Prime factorization of 3 = 3.

Prime factorization of 7 = 7.

Answer: LCM = 21.

LCM of 5, 8

Prime factorization of 5 = 5.

Prime factorization of 8 = 2 x 2 x 2.

Answer: LCM = 40.

LCM of 4, 10, 16

Prime factorization of 4 = 2 x 2.

Prime factorization of 10 = 2 x 5.

Prime factorization of 16 = 2 x 2 x 2 x 2.

Answer: LCM = 80.

LCM of 20, 25, 50

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 25 = 5 x 5.

Prime factorization of 50 = 2 x 5 x 5.

Answer: LCM = 100.

LCM of 45, 90, 95

Prime factorization of 45 = 3 x 3 x 5.

Prime factorization of 90 = 2 x 3 x 3 x 5.

Prime factorization of 95 = 5 x 19.

Answer: LCM = 1 710.

LCM of 32, 70, 80

Prime factorization of 32 = 2 x 2 x 2 x 2 x 2.

Prime factorization of 70 = 2 x 5 x 7.

Prime factorization of 80 = 2 x 2 x 2 x 2 x 5.

Answer: LCM = 1 120.

LCM of 33, 66

Prime factorization of 33 = 3 x 11.

Prime factorization of 66 = 2 x 3 x 11.

Answer: LCM = 66.

Q5. Find the LCM of the following numbers using division:

LCM of 4, 9

Prime factorization of 4 = 2 x 2.

Prime factorization of 9 = 3 x 3.

Answer: LCM = 36.

LCM of 7, 11

Prime factorization of 7 = 7.

Prime factorization of 11 = 11.

Answer: LCM = 77.

LCM of 14, 26

Prime factorization of 14 = 2 x 7.

Prime factorization of 26 = 2 x 13.

Answer: LCM = 182.

LCM of 20, 40

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 40 = 2 x 2 x 2 x 5.

Answer: LCM = 40.

LCM of 6, 24, 42

Prime factorization of 6 = 2 x 3.

Prime factorization of 24 = 2 x 2 x 2 x 3.

Prime factorization of 42 = 2 x 3 x 7.

Answer: LCM = 168.

LCM of 10, 20, 30

Prime factorization of 10 = 2 x 5.

Prime factorization of 20 = 2 x 2 x 5.

Prime factorization of 30 = 2 x 3 x 5.

Answer: LCM = 60.

LCM of 12, 18, 38

Prime factorization of 12 = 2 x 2 x 3.

Prime factorization of 18 = 2 x 3 x 3.

Prime factorization of 38 = 2 x 19.

Answer: LCM = 684.

LCM of 6, 15, 21

Prime factorization of 6 = 2 x 3.

Prime factorization of 15 = 3 x 5.

Prime factorization of 21 = 3 x 7.

Answer: LCM = 210.

Q6. 84 apples, 56 bananas and 24 oranges were distributed equally among some children. If the same combination of all kinds of fruits is distributed among all the children, find out the maximum possible number of children who can get the fruits?

Prime factorization of 84 = 2 x 2 x 3 x 7.

Prime factorization of 56 = 2 x 2 x 2 x 7.

Prime factorization of 24 = 2 x 2 x 2 x 3.

Common prime factors: 2 x 2 = 4.

Answer: HCF = 4.

Maximum possible number of children = 4.

Q7. Three water containers contain 12 litres, 24 litres and 42 litres of water. (a) Find the maximum capacity of a measuring container that can fully measure the quantity of water in all three containers. (b) Find out how many times this container needs to be filled to empty each container?

(a) Prime factorization of 12 = 2 x 2 x 3.

Prime factorization of 24 = 2 x 2 x 2 x 3.

Prime factorization of 42 = 2 x 3 x 7.

Common prime factors: 2 x 3 = 6.

Answer: HCF = 6.

Maximum capacity = 6 litres.

(b) 12 / 6 = 2 times, 24 / 6 = 4 times, 42 / 6 = 7 times.

Q8. Find the smallest number that can completely be divided by 32 and 55 without leaving any remainder.

Prime factorization of 32 = 2 x 2 x 2 x 2 x 2.

Prime factorization of 55 = 5 x 11.

Answer: LCM = 1 760.

Q9. Find the least number of stickers which can equally be distributed among 15, 12 and 10 children.

Prime factorization of 15 = 3 x 5.

Prime factorization of 12 = 2 x 2 x 3.

Prime factorization of 10 = 2 x 5.

Answer: LCM = 60.

Least number of stickers = 60.

## Unit 3: Fractions

### Exercise 1 - Addition and Subtraction of Fractions (book pages 41-46)

Worked example (book page 41): Yesterday we spent 1/2 hour and today we spent 3/4 hour in the computer lab. How much time have we spent altogether?

LCM of denominators = 4.

1/2 = 2/4

3/4 = 3/4

Add: 2/4 + 3/4 = 5/4 = 1 1/4.

Total time spent = 1 1/4 hours.

Worked example (book page 43): Sidra used 1 3/4 metre red ribbon and 7/8 metre blue ribbon to decorate the gift boxes. Find the total length of the ribbon she used.

Convert the mixed fraction into improper fraction: 1 3/4 = 7/4.

LCM of denominators = 8.

7/4 = 14/8

7/8 = 7/8

Add: 14/8 + 7/8 = 21/8 = 2 5/8.

Sidra used 2 5/8 m ribbon.

Worked example (book page 43): Sara uses 1/2 spoon sugar and 1/3 spoon of tea to make a cup of tea. Find how much more sugar she uses than the tea.

LCM of denominators = 6.

1/2 = 3/6, 1/3 = 2/6.

Subtract: 3/6 - 2/6 = 1/6 = 1/6.

Sara uses 1/6 spoon more sugar.

Try Yourself (book page 42): Observe the given shapes. Write the addition sentence for them and solve. (Add the shaded parts of the shapes as fractions with the same denominator.)

Q1. Solve the following fractions (addition):

(a) 1/2 + 1/2

LCM of denominators = 2.

1/2 = 1/2

1/2 = 1/2

Add: 1/2 + 1/2 = 2/2 = 1.

(b) 5 2/3 + 2 5/7

LCM of denominators = 21.

17/3 = 119/21

19/7 = 57/21

Add: 119/21 + 57/21 = 176/21 = 8 8/21.

(c) 3 4/5 + 5/7

LCM of denominators = 35.

19/5 = 133/35

5/7 = 25/35

Add: 133/35 + 25/35 = 158/35 = 4 18/35.

(d) 4 7/10 + 3/4

LCM of denominators = 20.

47/10 = 94/20

3/4 = 15/20

Add: 94/20 + 15/20 = 109/20 = 5 9/20.

(e) 7/9 + 3/4 + 2

LCM of denominators = 36.

7/9 = 28/36

3/4 = 27/36

2 = 72/36

Add: 28/36 + 27/36 + 72/36 = 127/36 = 3 19/36.

(f) 1 3/10 + 6 7/10 + 2 3/8

LCM of denominators = 40.

13/10 = 52/40

67/10 = 268/40

19/8 = 95/40

Add: 52/40 + 268/40 + 95/40 = 415/40 = 10 3/8.

(g) 4 + 2 7/12 + 1 19/24

LCM of denominators = 24.

4 = 96/24

31/12 = 62/24

43/24 = 43/24

Add: 96/24 + 62/24 + 43/24 = 201/24 = 8 3/8.

(h) 7/8 + 4 1/4 + 15/16

LCM of denominators = 16.

7/8 = 14/16

17/4 = 68/16

15/16 = 15/16

Add: 14/16 + 68/16 + 15/16 = 97/16 = 6 1/16.

Q2. Solve the following fractions (subtraction):

(a) 1 1/2 - 1/12

LCM of denominators = 12.

3/2 = 18/12, 1/12 = 1/12.

Subtract: 18/12 - 1/12 = 17/12 = 1 5/12.

(b) 2 8/9 - 1 2/3

LCM of denominators = 9.

26/9 = 26/9, 5/3 = 15/9.

Subtract: 26/9 - 15/9 = 11/9 = 1 2/9.

(c) 3 5/14 - 1 2/21

LCM of denominators = 42.

47/14 = 141/42, 23/21 = 46/42.

Subtract: 141/42 - 46/42 = 95/42 = 2 11/42.

(d) 5/6 - 6/11

LCM of denominators = 66.

5/6 = 55/66, 6/11 = 36/66.

Subtract: 55/66 - 36/66 = 19/66 = 19/66.

(e) 4 1/6 - 17/18

LCM of denominators = 18.

25/6 = 75/18, 17/18 = 17/18.

Subtract: 75/18 - 17/18 = 58/18 = 3 2/9.

(f) 1 3/4 - 4/5

LCM of denominators = 20.

7/4 = 35/20, 4/5 = 16/20.

Subtract: 35/20 - 16/20 = 19/20 = 19/20.

(g) 2 13/24 - 2/9

LCM of denominators = 72.

61/24 = 183/72, 2/9 = 16/72.

Subtract: 183/72 - 16/72 = 167/72 = 2 23/72.

(h) 5 1/8 - 1/3

LCM of denominators = 24.

41/8 = 123/24, 1/3 = 8/24.

Subtract: 123/24 - 8/24 = 115/24 = 4 19/24.

Q3. To practice for a marathon, Raheel ran 1/4 km on Monday, 7/8 km on Tuesday and 15/6 km on Wednesday. Find out how many kilometres did he run in three days?

LCM of denominators = 8.

1/4 = 2/8

7/8 = 7/8

5/2 = 20/8

Add: 2/8 + 7/8 + 20/8 = 29/8 = 3 5/8.

Raheel ran 3 5/8 km in three days.

Q4. Saad spent 2 1/2 hours for the preparation of Mathematics test and 1 1/4 hours for the preparation of Urdu test. Find: (a) in which subject did he spend more time and by how much? (b) how much time did he spend altogether?

(a) More time in Mathematics:

LCM of denominators = 4.

5/2 = 10/4, 5/4 = 5/4.

Subtract: 10/4 - 5/4 = 5/4 = 1 1/4.

(b) Total time:

LCM of denominators = 4.

5/2 = 10/4

5/4 = 5/4

Add: 10/4 + 5/4 = 15/4 = 3 3/4.

Saad spent 1 1/4 hours more in Mathematics, and 3 3/4 hours altogether.

Q5. An electrician has 18 8/9 m of wire. If he uses 2/9 m and 2 1/3 m in two rooms, then find: (a) how much wire did he use altogether? (b) how much wire is left?

(a) Wire used:

LCM of denominators = 9.

2/9 = 2/9

7/3 = 21/9

Add: 2/9 + 21/9 = 23/9 = 2 5/9.

(b) Wire left:

LCM of denominators = 9.

170/9 = 170/9, 23/9 = 23/9.

Subtract: 170/9 - 23/9 = 147/9 = 16 1/3.

He used 2 5/9 m wire altogether, and 16 1/3 m wire is left.

## Exercise 2 - Multiplication and Division of Fractions (book pages 47-52)

Worked example (book page 46): Ibrahim takes 3/4 hours daily to complete his homework. How much time does he spend on his homework in a week?

3/4 x 7 = 21/4 = 5 1/4 hours.

Ibrahim spends 5 1/4 hours on homework in a week.

Worked example (book page 47): Rabia completed one round of the park and covered a distance of 2 1/2 km. If she took 3 rounds of the park, how much distance did she cover?

2 1/2 x 3 = 5/2 x 3 = 15/2 = 7 1/2 km.

Rabia covered 7 1/2 km in 3 rounds.

Worked example (book page 49): For an experiment, I have to pour 7 litres of water in glasses with a capacity of 1/3 litres. Find the number of glasses to fill this quantity of water.

7 / (1/3) = 7 x 3 = 21 glasses.

I will need 21 glasses.

Q1. Solve the following and represent with figures:

(a) 1/2 x 12 (represent with figures)

1/2 x 12 = 6 = 6. (Draw 12 shapes, shade 1/2 of each, and count the shaded parts.)

(b) 5/9 x 7 (represent with figures)

5/9 x 7 = 35/9 = 3 8/9. (Draw 7 shapes, shade 5/9 of each, and count the shaded parts.)

(c) 6/7 x 3 (represent with figures)

6/7 x 3 = 18/7 = 2 4/7. (Draw 3 shapes, shade 6/7 of each, and count the shaded parts.)

(d) 3/4 x 4 (represent with figures)

3/4 x 4 = 3 = 3. (Draw 4 shapes, shade 3/4 of each, and count the shaded parts.)

(e) 2/3 x 5 (represent with figures)

2/3 x 5 = 10/3 = 3 1/3. (Draw 5 shapes, shade 2/3 of each, and count the shaded parts.)

(f) 5/6 x 9 (represent with figures)

5/6 x 9 = 15/2 = 7 1/2. (Draw 9 shapes, shade 5/6 of each, and count the shaded parts.)

Q2. Solve the following (multiplication):

(a) 4/5 x 1/4

Multiply numerator by numerator and denominator by denominator: 4/5 x 1/4 = 1/5 = 1/5.

(b) 3/7 x 7/9

Multiply numerator by numerator and denominator by denominator: 3/7 x 7/9 = 1/3 = 1/3.

(c) 1 2/3 x 3 3/4

Multiply numerator by numerator and denominator by denominator: 5/3 x 15/4 = 25/4 = 6 1/4.

(d) 29/30 x 1/5 x 1/2

Multiply numerator by numerator and denominator by denominator: 29/30 x 1/5 x 1/2 = 29/300 = 29/300.

(e) 2/7 x 4/5 x 3/7

Multiply numerator by numerator and denominator by denominator: 2/7 x 4/5 x 3/7 = 24/245 = 24/245.

(f) 1 1/14 x 1/4 x 2/3

Multiply numerator by numerator and denominator by denominator: 15/14 x 1/4 x 2/3 = 5/28 = 5/28.

(g) 1 3/5 x 2 3/7 x 3 3/4

Multiply numerator by numerator and denominator by denominator: 8/5 x 17/7 x 15/4 = 102/7 = 14 4/7.

(h) 5 2/7 x 8 2/5 x 9 3/11

Multiply numerator by numerator and denominator by denominator: 37/7 x 42/5 x 102/11 = 22644/55 = 411 39/55.

(i) 9 1/9 x 10 1/3 x 5 1/2

Multiply numerator by numerator and denominator by denominator: 82/9 x 31/3 x 11/2 = 13981/27 = 517 22/27.

Q3. Solve the following (division):

(a) 1/3 / 2/9

Divide by multiplying with the reciprocal: 1/3 / 2/9 = 1/3 x 9/2 = 3/2 = 1 1/2.

(b) 3/25 / 1/5

Divide by multiplying with the reciprocal: 3/25 / 1/5 = 3/25 x 5 = 3/5 = 3/5.

(c) 1/12 / 7/20

Divide by multiplying with the reciprocal: 1/12 / 7/20 = 1/12 x 20/7 = 5/21 = 5/21.

(d) 2 29/30 / 4 1/5

Divide by multiplying with the reciprocal: 89/30 / 21/5 = 89/30 x 5/21 = 89/126 = 89/126.

(e) 5 5/9 / 9 6/7

Divide by multiplying with the reciprocal: 50/9 / 69/7 = 50/9 x 7/69 = 350/621 = 350/621.

(f) 6 15/16 / 7 1/2

Divide by multiplying with the reciprocal: 111/16 / 15/2 = 111/16 x 2/15 = 37/40 = 37/40.

(g) 2 5/6 / 4 2/9

Divide by multiplying with the reciprocal: 17/6 / 38/9 = 17/6 x 9/38 = 51/76 = 51/76.

(h) 3 1/2 / 4/9

Divide by multiplying with the reciprocal: 7/2 / 4/9 = 7/2 x 9/4 = 63/8 = 7 7/8.

(i) 1/7 / 2 6/7

Divide by multiplying with the reciprocal: 1/7 / 20/7 = 1/7 x 7/20 = 1/20 = 1/20.

Q4. If 5 1/2 m of cloth is used to stitch one dress, how much cloth will be used to stitch 7 such dresses?

5 1/2 x 7 = 11/2 x 7 = 77/2 = 38 1/2 m.

38 1/2 m cloth will be used for 7 dresses.

Q5. Hina ate 1/2 part of a cake. If there were 8 pieces of cake, how many pieces did Hina eat?

1/2 x 8 = 4.

Hina ate 4 pieces.

Q6. Nida had 3/12 of a pizza. She gave 1/8 of it to her friend Madeeha. Find out what part of the whole pizza did Madeeha get?

3/12 x 1/8 = 3/96 = 1/32.

Madeeha got 1/32 part of the whole pizza.

Q7. To decorate the classroom, Hassan used ribbons of two colours. The length of blue ribbon is 5 7/8 metres. The length of red ribbon is 2/3 of the length of blue ribbon. (a) What is the length of red ribbon? (b) Find the total length of both ribbons.

(a) Red ribbon = 2/3 x 47/8 = 94/24 = 47/12 = 3 11/12 m.

(b) Total length:

LCM of denominators = 24.

47/8 = 141/24

47/12 = 94/24

Add: 141/24 + 94/24 = 235/24 = 9 19/24.

Red ribbon = 3 11/12 m, total length = 9 19/24 m.

Q8. 44 1/6 kilogram sugar is to be packed in 4 5/12 kilogram packets. Find: (a) in how many packets sugar will be packed? (b) how much sugar will be there in 9 packets of mass 4 5/12 kilograms?

(a) Divide by multiplying with the reciprocal: 265/6 / 53/12 = 265/6 x 12/53 = 10 = 10.

Sugar will be packed in 10 packets.

(b) 9 x 53/12 = 477/12 = 159/4 = 39 3/4 kg.

9 packets contain 39 3/4 kg sugar.

## Review Exercise - Unit 3 (book pages 52-53)

Q1. (a) 1/3 + 2/5 = ___.

(i) 3/5

✓ (ii) 11/15

(iii) 3/8

(iv) 2/15

(b) 2/3 - 5/9 = ___.

(i) 3/9

(ii) 7/3

✓ (iii) 1/9

(iv) 7/9

(c) The product of 3/4 and 4/3 is ___.

(i) 1000

(ii) 10

✓ (iii) 1

(iv) 111

Q1. (d) Which figure is showing 5 x 1/3?

The figure with five circles, each with one-third shaded, shows 5 x 1/3.

Q1. (e) In ___, the order of the fractions does not affect the result.

(i) addition and subtraction

(ii) subtraction and divisions

(iii) multiplication and division

✓ (iv) addition and multiplication

Q2. Solve the following:

(a) 7/20 + 4 3/10

LCM of denominators = 20.

7/20 = 7/20

43/10 = 86/20

Add: 7/20 + 86/20 = 93/20 = 4 13/20.

(b) 8/5 + 19/15 + 5 4/30

LCM of denominators = 15.

8/5 = 24/15

19/15 = 19/15

77/15 = 77/15

Add: 24/15 + 19/15 + 77/15 = 120/15 = 8.

(c) 3 14/50 - 2 9/25

LCM of denominators = 25.

82/25 = 82/25, 59/25 = 59/25.

Subtract: 82/25 - 59/25 = 23/25 = 23/25.

(d) 1 16/44 / 4/11

Divide by multiplying with the reciprocal: 15/11 / 4/11 = 15/11 x 11/4 = 15/4 = 3 3/4.

(e) 2 6/31 x 62/24

Multiply numerator by numerator and denominator by denominator: 68/31 x 31/12 = 17/3 = 5 2/3.

(f) 1 4/7 + 2 13/28 + 3/4

LCM of denominators = 28.

11/7 = 44/28

69/28 = 69/28

3/4 = 21/28

Add: 44/28 + 69/28 + 21/28 = 134/28 = 4 11/14.

(g) 3 6/8 + 3 3/4

LCM of denominators = 4.

15/4 = 15/4

15/4 = 15/4

Add: 15/4 + 15/4 = 30/4 = 7 1/2.

(h) 1 3/9 x 25

Multiply numerator by numerator and denominator by denominator: 4/3 x 25 = 100/3 = 33 1/3.

(i) 6 2/3 / 3 1/12

Divide by multiplying with the reciprocal: 20/3 / 37/12 = 20/3 x 12/37 = 80/37 = 2 6/37.

Q3. Find 2 3/4 of 36.

2 3/4 x 36 = 11/4 x 36 = 99.

Q4. How many months will be there in 3/4 of a year?

3/4 x 12 = 9 months.

Q5. Add the product of 4/5 and 6 to the quotient of 4/45 / 4/5.

Product = 4/5 x 6 = 24/5.

Quotient = 4/45 / 4/5 = 4/45 x 5/4 = 1/9.

Sum:

LCM of denominators = 45.

24/5 = 216/45

1/9 = 5/45

Add: 216/45 + 5/45 = 221/45 = 4 41/45.

Answer = 4 41/45.

Q6. Sana prepared 5 6/7 litres of apple juice. She served 3 2/5 litres of juice to the guests. Find out how much juice was left?

LCM of denominators = 35.

41/7 = 205/35, 17/5 = 119/35.

Subtract: 205/35 - 119/35 = 86/35 = 2 16/35.

2 16/35 litres juice was left.

Q7. Ali's mother bought 3 1/2 kg of chicken, 1 1/2 kg of fish and 2 1/4 kg of mutton. Find out how many kilograms of meat did she buy altogether?

LCM of denominators = 4.

7/2 = 14/4

3/2 = 6/4

9/4 = 9/4

Add: 14/4 + 6/4 + 9/4 = 29/4 = 7 1/4.

She bought 7 1/4 kg meat altogether.

Q8. Umair exercises for 2 1/5 hours daily. How many hours will he exercise in 30 days?

2 1/5 x 30 = 11/5 x 30 = 66 hours.

Umair will exercise 66 hours in 30 days.

Q9. The length of a side of the square is 1 7/8 metres. If the perimeter of a square is 4 x length of a side, find the perimeter of the square. Also, verify your answer by adding this length four times.

Perimeter = 4 x 15/8 = 60/8 = 15/2 = 7 1/2 m.

Verification: 15/8 + 15/8 + 15/8 + 15/8 = 60/8 = 7 1/2 m.

Q10. The area of a room is 209 1/18 m^2. If the length of the room is 17 2/3 m, find the width of the room.

Width = Area / Length = 3763/18 / 53/3 = 3763/18 x 3/53 = 3763/318 = 11 5/6 m.

Width of the room = 11 5/6 m.

## Unit 4: Decimals and Percentages

### Exercise 1 - Comparing and Ordering Decimals (book pages 55-57)

Try Yourself (book page 55): Compare by using the correct symbol: (i) 0.30 [ ] 0.3 (ii) 1.02 [ ] 1.22 (iii) 1.31 [ ] 1.3 (iv) 5.13 [ ] 5.03.

(i) 0.30 = 0.3 (putting 0 at the right of a decimal does not change its value).

(ii) 1.02 < 1.22.

(iii) 1.31 > 1.3.

(iv) 5.13 > 5.03.

Try It (book page 57): Put a decimal point in 999, 463, 208 and 175. Also arrange them in ascending and descending order.

One way: 9.99, 4.63, 2.08, 1.75.

Ascending order: 1.75, 2.08, 4.63, 9.99.

Descending order: 9.99, 4.63, 2.08, 1.75.

Q1. Compare the following decimals using the correct symbol (>, < or =):

0.4 [ ] 0.5

0.4 < 0.5.

1.3 [ ] 1.6

1.3 < 1.6.

34.56 [ ] 35.66

34.56 < 35.66.

6.67 [ ] 6.69

6.67 < 6.69.

0.45 [ ] 0.45

0.45 = 0.45.

23.12 [ ] 51.31

23.12 < 51.31.

71.2 [ ] 71.02

71.2 > 71.02.

6.06 [ ] 6.1

6.06 < 6.1.

4.8 [ ] 4.80

4.8 = 4.80.

Q2. Write the following decimals in ascending order:

7.2, 12.4, 10.8, 9.4, 18.2

Ascending order: 7.2, 9.4, 10.8, 12.4, 18.2.

8.56, 8.06, 8.5, 8.52, 8.51, 8.05

Ascending order: 8.05, 8.06, 8.5, 8.51, 8.52, 8.56.

3.4, 3.54, 3.55, 3.45, 3.14, 3.04

Ascending order: 3.04, 3.14, 3.4, 3.45, 3.54, 3.55.

9.5, 9.05, 9.52, 9.55, 9.09, 9.45

Ascending order: 9.05, 9.09, 9.45, 9.5, 9.52, 9.55.

Q3. Write the following decimals in descending order:

2.51, 3.31, 2.47, 3.81, 3.01

Descending order: 3.81, 3.31, 3.01, 2.51, 2.47.

4.32, 4.68, 0.31, 4.23, 4.31

Descending order: 4.68, 4.32, 4.31, 4.23, 0.31.

6.21, 6.77, 6.31, 6.33, 6.60

Descending order: 6.77, 6.60, 6.33, 6.31, 6.21.

1.62, 1.03, 1.60, 1.66, 1.61

Descending order: 1.66, 1.62, 1.61, 1.60, 1.03.

Exercise 2 - Addition and Subtraction of Decimals (book pages 58-60)

Worked example (book page 58): Hassan ran 2.62 kilometres in the morning and 1.06 kilometres in the evening. How many kilometres did he cover altogether?

2.62 + 1.06 = 3.68 km.

Hassan covered 3.68 km altogether.

Worked example (book page 59): A newborn giraffe is about 1.8 metres tall and an adult giraffe is about 5.5 metres tall. Find the difference in their heights.

5.5 - 1.8 = 3.7 m.

The difference in their heights is 3.7 m.

Q1. Solve the following (addition):

4.131 + 83

Align the decimal points and add:

4.131 + 83 = 87.131.

3.211 + 4.860

Align the decimal points and add:

3.211 + 4.860 = 8.071.

3.211 + 89.02

Align the decimal points and add:

3.211 + 89.02 = 92.231.

2.001 + 75.46

Align the decimal points and add:

2.001 + 75.46 = 77.461.

47.04 + 5.007

Align the decimal points and add:

47.04 + 5.007 = 52.047.

2.341 + 5.914

Align the decimal points and add:

2.341 + 5.914 = 8.255.

1.781 + 9.111

Align the decimal points and add:

1.781 + 9.111 = 10.892.

11.94 + 3.876

Align the decimal points and add:

11.94 + 3.876 = 15.816.

3.562 + 1.210

Align the decimal points and add:

3.562 + 1.210 = 4.772.

1.831 + 21.24

Align the decimal points and add:

1.831 + 21.24 = 23.071.

5.008 + 30.44

Align the decimal points and add:

5.008 + 30.44 = 35.448.

69.25 + 3.100

Align the decimal points and add:

69.25 + 3.100 = 72.35.

Q2. Solve the following (subtraction):

9.410 - 2.392

Align the decimal points and subtract:

9.410 - 2.392 = 7.018.

11.45 - 2.86

Align the decimal points and subtract:

11.45 - 2.86 = 8.59.

3.238 - 2.376

Align the decimal points and subtract:

3.238 - 2.376 = 0.862.

6.488 - 2.109

Align the decimal points and subtract:

6.488 - 2.109 = 4.379.

3.80 - 3.09

Align the decimal points and subtract:

3.80 - 3.09 = 0.71.

5.341 - 2.864

Align the decimal points and subtract:

5.341 - 2.864 = 2.477.

6.554 - 1.309

Align the decimal points and subtract:

6.554 - 1.309 = 5.245.

5.555 - 4.201

Align the decimal points and subtract:

5.555 - 4.201 = 1.354.

1.999 - 1.088

Align the decimal points and subtract:

1.999 - 1.088 = 0.911.

2.018 - 1.740

Align the decimal points and subtract:

2.018 - 1.740 = 0.278.

4.99 - 3.400

Align the decimal points and subtract:

4.99 - 3.400 = 1.59.

12.34 - 11.90

Align the decimal points and subtract:

12.34 - 11.90 = 0.44.

Q3. Rohan spent Rs 65.33 on Monday and Rs 97.29 on Tuesday. How much amount did he spend in two days?

65.33 + 97.29 = 162.62.

Rohan spent Rs 162.62 in two days.

Q4. Muaaz covered a distance of 76.36 km in car and 55.45 km in train to meet his grandmother. In which vehicle did he travel more and how much more?

76.36 - 55.45 = 20.91 km.

Muaaz travelled more in the car, by 20.91 km.

Q5. Saad ran 5.13 km on the first day and 2.33 km on the second day. Find the total distance he covered in both days.

5.13 + 2.33 = 7.46 km.

Saad covered 7.46 km in both days.

Q6. A tailor had 15.25 metre cloth. He used 11.55 metre cloth in stitching two dresses. How much cloth was left with him?

15.25 - 11.55 = 3.70 m.

3.70 m cloth was left with him.

Q7. Nida needs 50.55 gram sugar for a cake and 28.5 gram sugar for a few cupcakes. Find: (a) how much sugar does she need altogether? (b) which recipe requires more sugar and how much more?

(a) 50.55 + 28.5 = 79.05 g.

(b) The cake requires more sugar: 50.55 - 28.5 = 22.05 g more.

Nida needs 79.05 g sugar altogether; the cake needs 22.05 g more sugar.

## Exercise 3 - Multiplication of Decimals (book pages 61-66)

Worked example (book page 62): A piece of wood is 0.85 cm thick. If 10, 100 and 1 000 such pieces are piled, what will be their total thickness?

Thickness of 10 pieces = 0.85 x 10 = 8.5 cm.

Thickness of 100 pieces = 0.85 x 100 = 85 cm.

Thickness of 1 000 pieces = 0.85 x 1 000 = 850 cm.

Total thickness = 8.5 + 85 + 850 = 943.5 cm.

Worked example (book page 63): The capacity of a juice pack is 1.5 litres. What will be the capacity of 7 such packs?

1.5 x 7 = 10.5 litres.

The capacity of 7 packs is 10.5 litres.

Worked example (book page 64): A bus covers a distance of 9.85 kilometres going from one town to another and then coming back. If it takes 56 such rounds in a week, find the total distance it covers.

9.85 x 56 = 551.60 km.

The bus covers 551.60 km in a week.

Try Yourself (book page 64): What is the product of 12.9, 6.32 and 0.06?

12.9 x 6.32 = 81.528.

81.528 x 0.06 = 4.89168.

Product = 4.89168.

Q1. Solve the following (multiply by 10, 100, 1 000):

1.59 x 10

Move the decimal point 1 place(s) to the right: 1.59 x 10 = 15.9.

2.48 x 100

Move the decimal point 2 place(s) to the right: 2.48 x 100 = 248.

4.37 x 1000

Move the decimal point 3 place(s) to the right: 4.37 x 1000 = 4370.

5.43 x 100

Move the decimal point 2 place(s) to the right: 5.43 x 100 = 543.

3.03 x 100

Move the decimal point 2 place(s) to the right: 3.03 x 100 = 303.

3.06 x 1000

Move the decimal point 3 place(s) to the right: 3.06 x 1000 = 3060.

3.54 x 100

Move the decimal point 2 place(s) to the right: 3.54 x 100 = 354.

8.73 x 10

Move the decimal point 1 place(s) to the right: 8.73 x 10 = 87.3.

3.103 x 100

Move the decimal point 2 place(s) to the right: 3.103 x 100 = 310.3.

4.72 x 10

Move the decimal point 1 place(s) to the right: 4.72 x 10 = 47.2.

3.42 x 1000

Move the decimal point 3 place(s) to the right: 3.42 x 1000 = 3420.

2.44 x 100

Move the decimal point 2 place(s) to the right: 2.44 x 100 = 244.

Q2. Solve the following (multiply by a whole number):

5.36 x 82

Multiply as whole numbers, then place the decimal point:

5.36 x 82 = 439.52.

1.71 x 12

Multiply as whole numbers, then place the decimal point:

1.71 x 12 = 20.52.

4.91 x 34

Multiply as whole numbers, then place the decimal point:

4.91 x 34 = 166.94.

3.54 x 18

Multiply as whole numbers, then place the decimal point:

3.54 x 18 = 63.72.

1.23 x 2

Multiply as whole numbers, then place the decimal point:

1.23 x 2 = 2.46.

5.49 x 23

Multiply as whole numbers, then place the decimal point:

5.49 x 23 = 126.27.

2.51 x 89

Multiply as whole numbers, then place the decimal point:

2.51 x 89 = 223.39.

3.05 x 80

Multiply as whole numbers, then place the decimal point:

3.05 x 80 = 244.

1.46 x 61

Multiply as whole numbers, then place the decimal point:

1.46 x 61 = 89.06.

9.17 x 11

Multiply as whole numbers, then place the decimal point:

9.17 x 11 = 100.87.

2.50 x 43

Multiply as whole numbers, then place the decimal point:

2.50 x 43 = 107.5.

4.13 x 86

Multiply as whole numbers, then place the decimal point:

4.13 x 86 = 355.18.

Q3. Solve the following (multiply decimals):

5.36 x 2.5

Multiply as whole numbers, then place the decimal point:

5.36 x 2.5 = 13.4.

2.82 x 2.3

Multiply as whole numbers, then place the decimal point:

2.82 x 2.3 = 6.486.

5.82 x 5.34

Multiply as whole numbers, then place the decimal point:

5.82 x 5.34 = 31.0788.

4.65 x 39.7

Multiply as whole numbers, then place the decimal point:

4.65 x 39.7 = 184.605.

2.66 x 5.02

Multiply as whole numbers, then place the decimal point:

2.66 x 5.02 = 13.3532.

6.38 x 2.12

Multiply as whole numbers, then place the decimal point:

6.38 x 2.12 = 13.5256.

1.42 x 8.89

Multiply as whole numbers, then place the decimal point:

1.42 x 8.89 = 12.6238.

4.16 x 1.87

Multiply as whole numbers, then place the decimal point:

4.16 x 1.87 = 7.7792.

2.57 x 2.54

Multiply as whole numbers, then place the decimal point:

2.57 x 2.54 = 6.5278.

9.28 x 2.22

Multiply as whole numbers, then place the decimal point:

9.28 x 2.22 = 20.6016.

3.19 x 6.98

Multiply as whole numbers, then place the decimal point:

3.19 x 6.98 = 22.2662.

5.24 x 7.23

Multiply as whole numbers, then place the decimal point:

5.24 x 7.23 = 37.8852.

Q4. A train covers 96.5 kilometres in an hour. How much distance will be covered by the train in 25 hours?

96.5 x 25 = 2412.5 km.

The train will cover 2412.5 km in 25 hours.

Q5. 25.5 metres of fabric is required to make a sofa cover. How many metres of fabric is needed to make: (a) 10 covers (b) 100 covers (c) 1 000 covers?

(a) 25.5 x 10 = 255 m.

(b) 25.5 x 100 = 2550 m.

(c) 25.5 x 1000 = 25500 m.

Q6. A cement block weighs 4.23 kg. What will be the weight of 12 such blocks?

4.23 x 12 = 50.76 kg.

12 blocks weigh 50.76 kg.

Q7. The price of a litre of petrol is Rs 103.8. What will be the price of 35 litres of petrol?

103.8 x 35 = 3633.

35 litres of petrol cost Rs 3633.

## Exercise 4 - Division of Decimals (book pages 67-70)

Worked example (book page 67): Hina wants to cut 82.8 cm wire in 10 equal pieces. What will be the length of each piece?

82.8 / 10 = 8.28 cm.

Each piece is 8.28 cm long.

Worked example (book page 68): Saba wants to put 7.75 kilogram of rice in 5 packets equally. Find the quantity of rice in each packet.

7.75 / 5 = 1.55 kg.

Each packet has 1.55 kg rice.

Worked example (book page 68): Faraz read 49.5 pages of a book in 11 hours.

Find the number of pages he read in 1 hour.

49.5 / 11 = 4.5 pages.

Faraz read 4.5 pages in an hour.

Worked example (book page 69): If 13.5 kilogram of sugar is to be packed in boxes, each box contains 1.5 kg of sugar. Find the number of required boxes.

13.5 / 1.5 = 9 boxes.

9 boxes are required.

Try Yourself (book page 68): Divide 6.24 by 3.

6.24 / 3 = 2.08.

Try Yourself (book page 69): Without converting into fractions, divide 18.9 by 6.3 and divide 8 by 0.4.

18.9 / 6.3 = 3.

8 / 0.4 = 20.

Q1. Solve the following (divide by 10, 100, 1 000):

41.8 / 10

Move the decimal point 1 place(s) to the left: 41.8 / 10 = 4.18.

4.59 / 100

Move the decimal point 2 place(s) to the left: 4.59 / 100 = 0.0459.

5.48 / 1000

Move the decimal point 3 place(s) to the left: 5.48 / 1000 = 0.00548.

6.430 / 100

Move the decimal point 2 place(s) to the left: 6.430 / 100 = 0.0643.

4.14 / 10

Move the decimal point 1 place(s) to the left: 4.14 / 10 = 0.414.

6.17 / 1000

Move the decimal point 3 place(s) to the left: 6.17 / 1000 = 0.00617.

2.43 / 100

Move the decimal point 2 place(s) to the left: 2.43 / 100 = 0.0243.

71.4 / 10

Move the decimal point 1 place(s) to the left: 71.4 / 10 = 7.14.

4.24 / 100

Move the decimal point 2 place(s) to the left: 4.24 / 100 = 0.0424.

Q2. Solve the following (divide by a whole number):

3.40 / 2

Divide:

3.40 / 2 = 1.7.

5.73 / 3

Divide:

5.73 / 3 = 1.91.

5.05 / 5

Divide:

5.05 / 5 = 1.01.

2.2 / 5

Divide:

2.2 / 5 = 0.44.

3.66 / 3

Divide:

3.66 / 3 = 1.22.

57.8 / 34

Divide:

57.8 / 34 = 1.7.

80.6 / 4

Divide:

80.6 / 4 = 20.15.

8.19 / 18

Divide:

8.19 / 18 = 0.455.

62.4 / 52

Divide:

62.4 / 52 = 1.2.

Q3. Solve the following (divide by a decimal):

2.92 / 1.6

Divide:

2.92 / 1.6 = 1.825.

7.7 / 5

Divide:

7.7 / 5 = 1.54.

4.37 / 1.9

Divide:

4.37 / 1.9 = 2.3.

6.24 / 7.8

Divide:

6.24 / 7.8 = 0.8.

2.73 / 1.3

Divide:

2.73 / 1.3 = 2.1.

8.91 / 3.3

Divide:

8.91 / 3.3 = 2.7.

16.1 / 4.6

Divide:

16.1 / 4.6 = 3.5.

3.68 / 0.4

Divide:

3.68 / 0.4 = 9.2.

9.24 / 1.1

Divide:

9.24 / 1.1 = 8.4.

Q4. In 5 bags, 3.75 kg of flour is to be packed. How much flour will be in each bag?

3.75 / 5 = 0.75 kg.

Each bag has 0.75 kg flour.

Q5. Sana wants to fill 67.5 litres of oil in bottles. Find the number of bottles if (a) the capacity of the bottle is 4.5 litres (b) the capacity of the bottle is 2.5 litres.

(a) 67.5 / 4.5 = 15 bottles.

(b) 67.5 / 2.5 = 27 bottles.

Exercise 5 - Conversion of Fractions into Decimals (book page 72)

Worked example (book page 71): The swimming speed of a sea-turtle is 7/5 km per hour. How can we write this speed in decimals?

7 / 5 = 1.4 km per hour.

Worked example (book page 71): Grass has been grown on 3/4 part of a park. How can we write this as decimals?

3 / 4 = 0.75.

Q1. Convert the following fractions into decimals:

45/2

45 / 2 = 22.5.

19/25

19 / 25 = 0.76.

7/40

7 / 40 = 0.175.

33/100

33 / 100 = 0.33.

97/8

97 / 8 = 12.125.

78/8

78 / 8 = 9.75.

34/25

34 / 25 = 1.36.

2/5

2 / 5 = 0.4.

55/16

55 / 16 = 3.4375.

69/30

69 / 30 = 2.3.

72/45

72 / 45 = 1.6.

99/55

99 / 55 = 1.8.

Q2. Noman ate 1/8 of a pizza. How can we write this in decimal?

1 / 8 = 0.125.

Exercise 6 - Rounding-off and Estimating Decimals (book pages 73-74)

Worked example (book page 72): School library is at the distance of 1.47 km from the laboratory. Write this distance in kilometres to the nearest tenths.

In 1.47, the digit next to 4 is 7, which is greater than 5, so 1.47 = 1.5 (to the nearest tenth).

Worked example (book page 73): Umar's home is at a distance of 12.8 km from his office. The distance from the office to the bank is 20.3 km. First estimate the total distance and then verify your answer.

Estimate: 12.8 = 13, 20.3 = 20, so 13 + 20 = 33 km.

Actual: 12.8 + 20.3 = 33.1 km.

The estimation is correct.

Worked example (book page 74): Estimate the difference between 25.3 and 7.9, then verify.

Estimate: 25.3 = 25, 7.9 = 8, so 25 - 8 = 17.

Actual: 25.3 - 7.9 = 17.4.

The estimation is correct.

Try Yourself (book page 74): Estimate the sum and difference of the given numbers and then verify your answer: (a) 3.7, 11.4 (b) 30.8, 4.2.

(a) Sum estimate: 4 + 11 = 15, actual 3.7 + 11.4 = 15.1. Difference estimate: 11 - 4 = 7, actual 11.4 - 3.7 = 7.7.

(b) Sum estimate: 31 + 4 = 35, actual 30.8 + 4.2 = 35.0. Difference estimate: 31 - 4 = 27, actual 30.8 - 4.2 = 26.6.

Q1. Round-off the following decimals to one and two decimal places:

47.125

To one decimal place (nearest tenth): 47.1. To two decimal places (nearest hundredth): 47.13.

4.732

To one decimal place (nearest tenth): 4.7. To two decimal places (nearest hundredth): 4.73.

2.322

To one decimal place (nearest tenth): 2.3. To two decimal places (nearest hundredth): 2.32.

0.942

To one decimal place (nearest tenth): 0.9. To two decimal places (nearest hundredth): 0.94.

45.675

To one decimal place (nearest tenth): 45.7. To two decimal places (nearest hundredth): 45.68.

2.150

To one decimal place (nearest tenth): 2.2. To two decimal places (nearest hundredth): 2.15.

91.547

To one decimal place (nearest tenth): 91.5. To two decimal places (nearest hundredth): 91.55.

94.472

To one decimal place (nearest tenth): 94.5. To two decimal places (nearest hundredth): 94.47.

54.83

To one decimal place (nearest tenth): 54.8. To two decimal places (nearest hundredth): 54.83.

3.767

To one decimal place (nearest tenth): 3.8. To two decimal places (nearest hundredth): 3.77.

4.172

To one decimal place (nearest tenth): 4.2. To two decimal places (nearest hundredth): 4.17.

90.267

To one decimal place (nearest tenth): 90.3. To two decimal places (nearest hundredth): 90.27.

Q2. Estimate the sum of the given numbers. Then verify your answer.

52.90 + 47.98

Estimate: 53 + 48 = 101.

Actual: 52.90 + 47.98 = 100.88.

630.1 + 280.9

Estimate: 630 + 281 = 911.

Actual: 630.1 + 280.9 = 911.

41.01 + 36.87

Estimate: 41 + 37 = 78.

Actual: 41.01 + 36.87 = 77.88.

307.2 + 357.6

Estimate: 307 + 358 = 665.

Actual: 307.2 + 357.6 = 664.8.

747.2 + 30.10

Estimate: 747 + 30 = 777.

Actual: 747.2 + 30.10 = 777.3.

845.1 + 396.9

Estimate: 845 + 397 = 1242.

Actual: 845.1 + 396.9 = 1242.

63.81 + 25.91

Estimate: 64 + 26 = 90.

Actual: 63.81 + 25.91 = 89.72.

21.35 + 83.05

Estimate: 21 + 83 = 104.

Actual: 21.35 + 83.05 = 104.4.

99.99 + 87.91

Estimate: 100 + 88 = 188.

Actual: 99.99 + 87.91 = 187.9.

943.6 + 834.6

Estimate: 944 + 835 = 1779.

Actual: 943.6 + 834.6 = 1778.2.

123.4 + 567.8

Estimate: 123 + 568 = 691.

Actual: 123.4 + 567.8 = 691.2.

737.8 + 721.2

Estimate: 738 + 721 = 1459.

Actual: 737.8 + 721.2 = 1459.

Q3. Estimate the difference of the given numbers. Then verify your answer.

22.30 - 17.99

Estimate: 22 - 18 = 4.

Actual: 22.30 - 17.99 = 4.31.

78.92 - 69.11

Estimate: 79 - 69 = 10.

Actual: 78.92 - 69.11 = 9.81.

56.23 - 11.26

Estimate: 56 - 11 = 45.

Actual: 56.23 - 11.26 = 44.97.

234.6 - 159.8

Estimate: 235 - 160 = 75.

Actual: 234.6 - 159.8 = 74.8.

587.6 - 320.9

Estimate: 588 - 321 = 267.

Actual: 587.6 - 320.9 = 266.7.

402.3 - 292.1

Estimate: 402 - 292 = 110.

Actual: 402.3 - 292.1 = 110.2.

995.5 - 747.1

Estimate: 996 - 747 = 249.

Actual: 995.5 - 747.1 = 248.4.

673.1 - 430.5

Estimate: 673 - 431 = 242.

Actual: 673.1 - 430.5 = 242.6.

53.25 - 25.62

Estimate: 53 - 26 = 27.

Actual: 53.25 - 25.62 = 27.63.

54.41 - 41.45

Estimate: 54 - 41 = 13.

Actual: 54.41 - 41.45 = 12.96.

3.5 - 2.1

Estimate: 4 - 2 = 2.

Actual: 3.5 - 2.1 = 1.4.

9.355 - 7.316

Estimate: 9 - 7 = 2.

Actual: 9.355 - 7.316 = 2.039.

Exercise 7 - Percentages (book pages 75-79)

Worked example (book page 75): Mariyam got 23 marks out of 25. Calculate the percentage of her marks?

23/25 = (23 x 4)/(25 x 4) = 92/100 = 92%.

Mariyam's percentage is 92%.

Worked example (book page 77): In an annual examination of Social Studies, 30% questions are based on mapping skills. If there are 20 questions altogether, find the number of questions based on mapping skills.

30% of 20 = (30/100) x 20 = 6.

6 questions are based on mapping skills.

Worked example (book page 78): A factory produced 850 masks in a day. 72% masks are blue. Find the number of blue masks produced by the factory in a day.

72% of 850 = (72/100) x 850 = 612.

The factory produced 612 blue masks in a day.

Try It (book page 78): 20% of an amount is 350 rupees. Find: (i) 1% of the amount (ii) 100% of the amount (iii) 1 000% of the amount.

(i) 1% = 350 / 20 = 17.5 rupees.

(ii) 100% = 17.5 x 100 = 1750 rupees.

(iii) 1 000% = 17.5 x 1000 = 17500 rupees.

Try Yourself (book page 78): What is 62% of 62? What is 5% of 50?

62% of 62 = (62/100) x 62 = 38.44.

5% of 50 = (5/100) x 50 = 2.5.

Q1. Express the following in percentage:

4/100

4/100 = 4%.

35/100

35/100 = 35%.

76/100

76/100 = 76%.

12/100

12/100 = 12%.

28/100

28/100 = 28%.

47/100

47/100 = 47%.

45/100

45/100 = 45%.

66/100

66/100 = 66%.

89/100

89/100 = 89%.

1/100

1/100 = 1%.

Q2. Convert the following percentages into fractions:

13%

13% = 13/100 = 13/100 (lowest form).

24%

24% = 24/100 = 6/25 (lowest form).

46%

46% = 46/100 = 23/50 (lowest form).

55%

55% = 55/100 = 11/20 (lowest form).

68%

68% = 68/100 = 17/25 (lowest form).

72%

72% = 72/100 = 18/25 (lowest form).

87%

87% = 87/100 = 87/100 (lowest form).

98%

98% = 98/100 = 49/50 (lowest form).

11%

11% = 11/100 = 11/100 (lowest form).

10%

10% = 10/100 = 1/10 (lowest form).

Q3. Convert the following percentages into decimals:

15%

15% = 15/100 = 0.15.

26%

26% = 26/100 = 0.26.

47%

47% = 47/100 = 0.47.

52%

52% = 52/100 = 0.52.

63%

63% = 63/100 = 0.63.

74%

74% = 74/100 = 0.74.

85%

85% = 85/100 = 0.85.

96%

96% = 96/100 = 0.96.

17%

17% = 17/100 = 0.17.

18%

18% = 18/100 = 0.18.

Q4. Convert the following fractions into percentage:

1/10

1/10 = 10%.

2/5

2/5 = 40%.

4/10

4/10 = 40%.

9/25

9/25 = 36%.

3/20

3/20 = 15%.

21/50

21/50 = 42%.

1/5

1/5 = 20%.

7/20

7/20 = 35%.

9/10

9/10 = 90%.

6/25

6/25 = 24%.

Q5. Convert the following decimals into percentage:

0.06

0.06 x 100 = 6%.

0.14

0.14 x 100 = 14%.

0.23

0.23 x 100 = 23%.

0.34

0.34 x 100 = 34%.

0.43

0.43 x 100 = 43%.

0.55

0.55 x 100 = 55%.

0.63

0.63 x 100 = 63%.

0.71

0.71 x 100 = 71%.

0.98

0.98 x 100 = 98%.

0.3

0.3 x 100 = 30%.

Q6. Sajid scored 365 marks out of 500. What percentage did he score?

365/500 x 100 = 73%.

Sajid scored 73%.

Q7. There are 40 students in a class. If 5% are absent on Monday, find: (a) the number of absent students (b) the number of present students.

(a) 5% of 40 = (5/100) x 40 = 2 students absent.

(b) 40 - 2 = 38 students present.

Q8. In an exam, 450 students appeared. 25% got first division, 55% got second division and the rest just passed the exam. Find the number of students who just passed the exam.

Rest = 100% - 25% - 55% = 20%.

20% of 450 = (20/100) x 450 = 90 students.

90 students just passed the exam.

Q9. 12% of apples in a basket are red. Write this percentage in fraction and decimal.

Fraction: 12% = 12/100 = 3/25.

Decimal: 12% = 0.12.

Q10. A tank can hold 85 litres of water. It is 30% filled. What percentage of the tank is empty? Write the answer in decimal and fraction.

Empty = 100% - 30% = 70%.

Decimal: 70% = 0.70.

Fraction: 70% = 70/100 = 7/10.

Review Exercise - Unit 4 (book pages 80-82)

Q1. (a) While putting ___ at the right of a decimal does not affect its value.

(i) 100

✓ (ii) 0

(iii) 4

(iv) 10

(b) When we multiply a decimal by 100, we move the decimal point 2 places to the ___.

(i) left

(ii) down

(iii) up

✓ (iv) right

(c) We represent percentage by the symbol ___.

(i) <

(ii) >

(iii) +

(iv) % ✓

(d) 20% of 540 is ___.

(i) 37

(ii) 108 ✓

(iii) 27

(iv) 270

Q1. (e) Percentage is a special kind of ___.

Percentage is a special kind of fraction (with 100 as denominator).

Q2. Compare the following decimals using the correct symbol (>, < or =):

0.05 [ ] 0.8

0.05 < 0.8.

1.8 [ ] 1.4

1.8 > 1.4.

45.67 [ ] 45.77

45.67 < 45.77.

7.0 [ ] 7.0

7.0 = 7.0.

1.56 [ ] 1.56

1.56 = 1.56.

34.23 [ ] 62.42

34.23 < 62.42.

Q3. Solve the following:

5.242 + 9.003

Align the decimal points and add:

5.242 + 9.003 = 14.245.

3.622 + 22.971

Align the decimal points and add:

3.622 + 22.971 = 26.593.

4.32 + 90.16

Align the decimal points and add:

4.32 + 90.16 = 94.48.

13.12 + 86.57

Align the decimal points and add:

13.12 + 86.57 = 99.69.

58.57 + 6.118

Align the decimal points and add:

58.57 + 6.118 = 64.688.

45.61 + 27.16

Align the decimal points and add:

45.61 + 27.16 = 72.77.

92.93 - 31.33

Align the decimal points and subtract:

92.93 - 31.33 = 61.6.

8.25 - 4.97

Align the decimal points and subtract:

8.25 - 4.97 = 3.28.

4.63 - 1.21

Align the decimal points and subtract:

4.63 - 1.21 = 3.42.

22.92 - 2.001

Align the decimal points and subtract:

22.92 - 2.001 = 20.919.

6.119 - 1.55

Align the decimal points and subtract:

6.119 - 1.55 = 4.569.

20.36 - 6.211

Align the decimal points and subtract:

20.36 - 6.211 = 14.149.

Q4. Ahmed wants to buy a chocolate which costs Rs 98.46. He has Rs 52.25. How much more money does he need to buy the chocolate?

98.46 - 52.25 = 46.21.

Ahmed needs Rs 46.21 more.

Q5. Omar bought two rolls of tape. The first roll has 16.38 metres of tape and the second roll has 56.82 metres of tape. Find: (a) how much tape was there in both of the rolls altogether? (b) which roll has more tape and how much?

(a) 16.38 + 56.82 = 73.20 m.

(b) The second roll has more tape: 56.82 - 16.38 = 40.44 m more.

Q6. In a fabric warehouse, there was 43.5 metres of coloured cloth. If it is cut into equal pieces of 1.5 metres, then: (a) how many pieces of cloth will be obtained? (b) what will be the total length of 12 pieces of cloth of length 1.5 metres?

(a) 43.5 / 1.5 = 29 pieces.

(b) 12 x 1.5 = 18 m.

Q7. Solve the following:

32.855 x 10

32.855 x 10 = 328.55.

4.39 x 100

4.39 x 100 = 439.

5.98 x 1000

5.98 x 1000 = 5980.

6.54 x 21

Multiply as whole numbers, then place the decimal point:

6.54 x 21 = 137.34.

4.14 x 43

Multiply as whole numbers, then place the decimal point:

4.14 x 43 = 178.02.

7.17 x 6.5

Multiply as whole numbers, then place the decimal point:

7.17 x 6.5 = 46.605.

69.7 x 2.31

Multiply as whole numbers, then place the decimal point:

69.7 x 2.31 = 161.007.

2.19 x 4.87

Multiply as whole numbers, then place the decimal point:

2.19 x 4.87 = 10.6653.

4.13 x 6.12

Multiply as whole numbers, then place the decimal point:

4.13 x 6.12 = 25.2756.

82.6 / 100

Divide:

82.6 / 100 = 0.826.

3.12 / 1.3

Divide:

3.12 / 1.3 = 2.4.

4.21 / 2.5

Divide:

4.21 / 2.5 = 1.684.

Q8. Estimate the sum and difference of the given numbers. Then verify your answer.

63.11 + 2.809 (estimate the sum, then verify)

Estimate: 63 + 3 = 66.

Actual: 63.11 + 2.809 = 65.919.

74.1 + 3.9 (estimate the sum, then verify)

Estimate: 74 + 4 = 78.

Actual: 74.1 + 3.9 = 78.

521.2 + 47 (estimate the sum, then verify)

Estimate: 521 + 47 = 568.

Actual: 521.2 + 47 = 568.2.

Q9. Round-off the following decimals to the nearest tenths and hundredths.

2.2342

Nearest tenth: 2.2. Nearest hundredth: 2.23.

3.1723

Nearest tenth: 3.2. Nearest hundredth: 3.17.

5.3671

Nearest tenth: 5.4. Nearest hundredth: 5.37.

9.5191

Nearest tenth: 9.5. Nearest hundredth: 9.52.

Q10. Complete the table (Fraction / Decimal / Percent).

(a) 21/50

21/50 = 0.42 = 42%.

(b) 82%

82% = 0.82 = 82/100 = 41/50.

(c) 7/25

7/25 = 0.28 = 28%.

(d) 0.65

0.65 = 65/100 = 13/20 = 65%.

(e) 25%

25% = 0.25 = 25/100 = 1/4.

Q11. Junaid spent Rs 432 out of Rs 600. What is the percentage of the total amount spent by Junaid? Write the answer in fraction and decimals.

432/600 x 100 = 72%.

Fraction: 72/100 = 18/25.

Decimal: 0.72.

Q12. There are 32 students in a class. If 25% students wear glasses, find: (a) the number of students who wear glasses (b) how many students do not wear glasses?

(a) 25% of 32 = (25/100) x 32 = 8 students wear glasses.

(b) 32 - 8 = 24 students do not wear glasses.

Q13. A Mathematics paper has 20 questions out of which 60% are based on 'Numbers and Operations'. (a) How many questions are based on 'Numbers and Operations'? (b) Write the number of remaining questions in fraction and decimal.

(a) 60% of 20 = (60/100) x 20 = 12 questions.

(b) Remaining questions = 20 - 12 = 8. As a fraction of the paper: 8/20 = 2/5. As a decimal: 0.4.

## Unit 5: Distance and Time

### Exercise 1 - Conversion of Units of Distance (book pages 84-91)

Worked example (book page 83): Farhan participated in a cycle race. He has to cover a distance of 15 kilometres to win the race. How many metres are there in 15 kilometres?

1 km = 1 000 m, so 15 km = 15 x 1 000 = 15 000 m.

Worked example (book page 84): Children are going to Murree from Islamabad for a recreational trip. They cover the distance of 55 km. How many metres do they cover?

55 km = 55 x 1 000 = 55 000 m.

Try Yourself (book page 84): Nanga Parbat is the ninth highest mountain in the world. Its height is 8 kilometres 126 metres. What will be its height in metres?

8 km 126 m = 8 x 1 000 m + 126 m = 8 000 m + 126 m = 8 126 m.

Worked example (book page 85): The distance of my home to the park is 2 000 metres. Find this distance in kilometres.

2 000 m = 2 000 / 1 000 = 2 km.

Worked example (book page 85): Convert 9 800 metres to kilometres and metres.

9 800 m = 9 000 m + 800 m = 9 km 800 m.

Try Yourself (book page 85): Convert the following distances into kilometres and metres: (i) 5 000 m (ii) 2 002 m (iii) 8 976 m.

(i) 5 000 m = 5 km.

(ii) 2 002 m = 2 km 2 m.

(iii) 8 976 m = 8 km 976 m.

Worked example (book page 86): The height of Minar-e-Pakistan is 70 m. What will be its height in centimetres?

70 m = 70 x 100 = 7 000 cm.

Try Yourself (book page 86): A tree is 7.5 metres high. What will be its height in centimetres?

7.5 m = 7.5 x 100 = 750 cm.

Worked example (book page 86): Convert 38 m 78 cm in centimetres only.

38 m 78 cm = 38 x 100 cm + 78 cm = 3 800 cm + 78 cm = 3 878 cm.

Worked example (book page 87): The height of a cupboard is 200 centimetres. What will be its height in metres?

200 cm = 200 / 100 = 2 m.

Worked example (book page 87): Convert 585 cm into metres and centimetres.

585 cm = 500 cm + 85 cm = 5 m 85 cm.

Try Yourself (book page 87): Convert the following distances into metres: (i) 670 cm (ii) 409 cm (iii) 900 cm.

(i) 670 cm = 6 m 70 cm.

(ii) 409 cm = 4 m 9 cm.

(iii) 900 cm = 9 m.

Worked example (book page 88): The length of Anmer's lunch box is 15 cm. What will be its length in millimetres?

15 cm = 15 x 10 = 150 mm.

Try Yourself (book page 88): Ahmed buys a toy car which is 24 cm long. What will be its length in millimetres?

24 cm = 24 x 10 = 240 mm.

Worked example (book page 88): Convert 89 centimetres 44 millimetres into millimetres.

89 cm 44 mm = 89 x 10 mm + 44 mm = 890 mm + 44 mm = 934 mm.

Worked example (book page 89): I bought a picture frame. Its length is 60 millimetre. Find its length in centimetres.

60 mm = 60 / 10 = 6 cm.

Worked example (book page 89): Convert 18 millimetres into centimetres and millimetres.

18 mm = 10 mm + 8 mm = 1 cm 8 mm.

Try Yourself (book page 89): Convert the following distances into centimetres: (i) 20 mm (ii) 78 mm (iii) 98 mm.

(i) 20 mm = 2 cm.

(ii) 78 mm = 7 cm 8 mm.

(iii) 98 mm = 9 cm 8 mm.

## Exercise 2 - Addition and Subtraction of Distance (book pages 90-92)

Worked example (book page 90): Subhan walks 2 kilometres 670 metres on Saturday and 3 kilometres 490 metres on Sunday. What is the total distance he covers in metres in both the days?

2 km 670 m + 3 km 490 m = 5 km 1 160 m = 6 km 160 m.

In metres: 6 km 160 m = 6 x 1 000 m + 160 m = 6 160 m.

Worked example (book page 91): Petronas Twin Towers in Malaysia are 452 m high, while Burj Khalifa in Dubai is 828 metres high. What is the difference between their heights?

828 m - 452 m = 376 m.

The difference is 376 m.

Try Yourself (book page 91): Solve the following: (i) 670 m - 450 m (ii) 83 km 219 m - 51 km 340 m (iii) 32 cm 3 mm - 9 cm 4 mm.

(i) 670 m - 450 m = 220 m.

(ii) 83 km 219 m - 51 km 340 m: borrow 1 km: 82 km 1 219 m - 51 km 340 m = 31 km 879 m.

(iii) 32 cm 3 mm - 9 cm 4 mm: borrow 1 cm: 31 cm 13 mm - 9 cm 4 mm = 22 cm 9 mm.

Q1. Convert the following units of distance as directed:

(a) 34 km into m

34 km = 34 x 1 000 = 34 000 m.

(b) 74 km 930 m into m

74 km 930 m = 74 x 1 000 m + 930 m = 74 930 m.

(c) 1 970 m into km and m

1 970 m = 1 km 970 m.

(d) 5 890 m into km and m

5 890 m = 5 km 890 m.

(e) 67 m into cm

67 m = 67 x 100 = 6 700 cm.

(f) 650 m 46 cm into cm

650 m 46 cm = 650 x 100 cm + 46 cm = 65 046 cm.

(g) 840 cm into m and cm

840 cm = 8 m 40 cm.

(h) 107 cm into mm

107 cm = 107 x 10 = 1 070 mm.

(i) 99 cm 6 mm into mm

99 cm 6 mm = 99 x 10 mm + 6 mm = 996 mm.

(j) 70 mm into cm

70 mm = 70 / 10 = 7 cm.

(k) 485 mm into cm and mm

485 mm = 48 cm 5 mm.

(l) 900 m into cm

900 m = 900 x 100 = 90 000 cm.

Q2. To celebrate Independence Day, Madeeha bought 4 m 35 cm green colour cloth to stitch a green shirt. For shawl and trouser, she bought 6 m 79 cm white colour cloth. How many centimetres of cloth did she buy altogether?

4 m 35 cm = 435 cm, 6 m 79 cm = 679 cm.

435 cm + 679 cm = 1 114 cm.

Madeeha bought 1 114 cm cloth altogether.

Q3. Ahmad bought 140 cm ribbon to pack the gift box. How many millimetre ribbon did he buy?

140 cm = 140 x 10 = 1 400 mm.

Ahmad bought 1 400 mm ribbon.

Q4. The lengths of the two ropes are 17 cm 9 mm and 80 cm 6 mm. (a) What is the difference between the lengths of the two ropes? (b) What is the total length of the two ropes in millimetres?

(a) 80 cm 6 mm - 17 cm 9 mm: borrow 1 cm: 79 cm 16 mm - 17 cm 9 mm = 62 cm 7 mm.

(b) 17 cm 9 mm = 179 mm, 80 cm 6 mm = 806 mm. Total = 179 + 806 = 985 mm.

Q5. In a hospital, two halls are constructed for patients. The length of one hall is 276 m 20 cm and the length of the other hall is 689 m 98 cm. What is the total length of both halls?

276 m 20 cm = 27 620 cm, 689 m 98 cm = 68 998 cm.

27 620 + 68 998 = 96 618 cm = 966 m 18 cm.

Total length = 966 m 18 cm.

Q6. The distance between Ahmer's house to Masjid is 4 km 196 m, the distance between Ali's house to Masjid is 5 km 298 m. Whose house is nearer to the Masjid and by how much?

Ahmer's house is nearer: 5 km 298 m - 4 km 196 m = 1 km 102 m.

Ahmer's house is nearer by 1 km 102 m.

Q7. The park near Waleed's house is 2 km 117 m long and the park near Maheen's house is 3 km 214 m long. What is the difference between the lengths of the two parks in metres?

3 km 214 m - 2 km 117 m = 1 km 97 m = 1 097 m.

The difference is 1 097 m.

## Exercise 3 - Conversion of Units of Time (book pages 92-101)

Worked example (book page 92): Farhan completes a science project in 4 hours. In how many minutes does he complete the project?

4 hr = 4 x 60 = 240 min.

Worked example (book page 93): Convert 12 hours 45 minutes into minutes.

12 hr 45 min = 12 x 60 min + 45 min = 720 + 45 = 765 min.

Worked example (book page 93): A doctor performs his duty for 480 minutes daily. How many hours does he work daily?

480 min = 480 / 60 = 8 hr.

Worked example (book page 94): Convert 5 280 minutes into hours.

5 280 min = 5 280 / 60 = 88 hr.

Try Yourself (book page 94): Convert the following minutes into hours: (i) 380 min (ii) 720 min (iii) 440 min.

(i) 380 min = 6 hr 20 min.

(ii) 720 min = 12 hr.

(iii) 440 min = 7 hr 20 min.

Worked example (book page 94): Aadil finishes his lunch in 25 minutes. In how many seconds does he finish his lunch?

25 min = 25 x 60 = 1 500 sec.

Worked example (book page 95): Convert 120 minutes 59 seconds into seconds.

120 min 59 sec = 120 x 60 sec + 59 sec = 7 200 + 59 = 7 259 sec.

Worked example (book page 95): To avoid Coronavirus, hands must be washed for at least 20 seconds. For how many minutes must the hands be washed?

20 sec = 20 / 60 = 0.33 min.

Worked example (book page 95): Convert 300 seconds into minutes.

300 sec = 300 / 60 = 5 min.

Try Yourself (book page 95): Convert the following into seconds: (i) 220 min (ii) 540 min (iii) 720 min.

(i) 220 min = 13 200 sec.

(ii) 540 min = 32 400 sec.

(iii) 720 min = 43 200 sec.

Worked example (book page 96): Convert 240 hours into days.

240 hr = 240 / 24 = 10 days.

Worked example (book page 96): Convert 5 days into hours.

5 days = 5 x 24 = 120 hr.

Worked example (book page 96): A gardener plants a seed. After 2 weeks, a small plant comes out. In how many days did the plant come out?

2 weeks = 2 x 7 = 14 days.

Worked example (book page 96): Convert 6 weeks 3 days into days.

6 weeks 3 days = 6 x 7 days + 3 days = 42 + 3 = 45 days.

Try Yourself (book page 97): A toy factory manufactures 568 toys in 12 weeks and 6 days. In how many days does it manufacture these toys?

12 weeks 6 days = 12 x 7 + 6 = 84 + 6 = 90 days.

Worked example (book page 97): Hamid goes to Islamabad for 21 days. How many weeks does he spend in Islamabad?

21 days = 21 / 7 = 3 weeks.

Worked example (book page 97): Convert 420 days into weeks.

420 days = 420 / 7 = 60 weeks.

Try Yourself (book page 97): February has 28 days except in the leap year. How many weeks are there in February?

28 days = 28 / 7 = 4 weeks.

Worked example (book page 98): Mubeen needs 11 months to construct a house. How much time will be required for construction in days?

11 months = 11 x 30 = 330 days.

Worked example (book page 98): Convert 12 months and 17 days into days.

12 months 17 days = 12 x 30 days + 17 days = 360 + 17 = 377 days.

Try Yourself (book page 98): Imran completes a storybook in 2 months. In how many days does he complete the storybook?

2 months = 2 x 30 = 60 days.

Worked example (book page 99): An author writes a book in 90 days. In how many months does he write the book?

90 days = 90 / 30 = 3 months.

Worked example (book page 99): Convert 660 days into months.

660 days = 660 / 30 = 22 months.

Try Yourself (book page 99): Convert the following into months: (i) 520 days (ii) 15 days (iii) 30 days.

(i) 520 days = 17 months 10 days (520 = 17 x 30 + 10).

(ii) 15 days = 15/30 = 1/2 month.

(iii) 30 days = 1 month.

Worked example (book page 100): Ahmad completes a project in 3 years. In how many months does he complete this project?

3 years = 3 x 12 = 36 months.

Worked example (book page 100): Convert 15 years 7 months into months.

15 years 7 months = 15 x 12 months + 7 months = 180 + 7 = 187 months.

Try Yourself (book page 100): Tahir works in a factory for 5 years. For how many months does he work in the factory?

5 years = 5 x 12 = 60 months.

Worked example (book page 100): The government is constructing homes for people and this project will be completed in 18 months. In how many years will the government complete the project?

18 months = 18 / 12 = 1.5 years.

Worked example (book page 101): Convert 120 months into years.

120 months = 120 / 12 = 10 years.

Try Yourself (book page 101): Ahmad is 18 years old. Find his age in months.

18 years = 18 x 12 = 216 months.

Q1. Convert the given units of time as directed:

(a) 45 hr into min

45 hr = 45 x 60 = 2 700 min.

(b) 240 hr 56 min into min

240 hr 56 min = 240 x 60 + 56 = 14 400 + 56 = 14 456 min.

(c) 960 min into hr

960 min = 960 / 60 = 16 hr.

(d) 440 min into hr and min

440 min = 7 hr 20 min.

(e) 64 min into sec

64 min = 64 x 60 = 3 840 sec.

(f) 180 min into sec

180 min = 180 x 60 = 10 800 sec.

(g) 544 sec into min and sec

544 sec = 9 min 4 sec.

(h) 600 sec into min

600 sec = 600 / 60 = 10 min.

Q2. Convert the following as directed:

(a) 56 years into months

56 years = 56 x 12 = 672 months.

(b) 34 years 10 months into months

34 years 10 months = 34 x 12 + 10 = 408 + 10 = 418 months.

(c) 48 months into years

48 months = 48 / 12 = 4 years.

(d) 56 months into years and months

56 months = 4 years 8 months.

(e) 78 weeks into days

78 weeks = 78 x 7 = 546 days.

(f) 12 weeks 6 days into days

12 weeks 6 days = 12 x 7 + 6 = 90 days.

(g) 49 days into weeks

49 days = 49 / 7 = 7 weeks.

(h) 180 days into months

180 days = 180 / 30 = 6 months.

(i) 67 months into days

67 months = 67 x 30 = 2 010 days.

(j) 44 months 29 days into days

44 months 29 days = 44 x 30 + 29 = 1 320 + 29 = 1 349 days.

## Exercise 4 - Addition and Subtraction of Units of Time (book pages 102-104)

Worked example (book page 102): Hadia bakes a cake in 45 minutes 28 seconds and a pizza in 55 minutes 34 seconds. In how many seconds does she bake both the items?

45 min 28 sec + 55 min 34 sec: seconds 28 + 34 = 62 sec = 1 min 2 sec.

Total = 100 min + 1 min 2 sec = 101 min 2 sec.

In seconds: 101 x 60 + 2 = 6 060 + 2 = 6 062 sec.

Worked example (book page 102): Ahmad spends 15 weeks 5 days in his grandmother's home and 6 weeks 6 days in his aunt's home. How many more days does he spend in his grandmother's home?

15 weeks 5 days - 6 weeks 6 days: borrow 1 week: 14 weeks 12 days - 6 weeks 6 days = 8 weeks 6 days.

In days: 8 x 7 + 6 = 56 + 6 = 62 days.

Ahmad spends 62 more days in his grandmother's home.

Try Yourself (book page 103): Subtract the given units of time: (i) 13 years - 12 years 6 months (ii) 8 hours 56 minutes - 7 hours 12 minutes.

(i) 13 years - 12 years 6 months = 6 months.

(ii) 8 hr 56 min - 7 hr 12 min = 1 hr 44 min.

Q1. Solve the following (addition of time):

(a) 3 hr 20 min + 5 hr 43 min

3 hr 20 min + 5 hr 43 min = 8 hr 63 min = 9 hr 3 min.

(b) 13 min 12 sec + 15 min 19 sec

13 min 12 sec + 15 min 19 sec = 28 min 31 sec.

(c) 33 years 8 months + 40 years 11 months

33 yr 8 mo + 40 yr 11 mo = 73 yr 19 mo = 74 yr 7 mo.

(d) 2 weeks 3 days + 8 weeks 1 day

2 wk 3 d + 8 wk 1 d = 10 wk 4 d.

(e) 117 months + 7 months

117 months + 7 months = 124 months.

(f) 8 months 12 days + 2 months 14 days

8 mo 12 d + 2 mo 14 d = 10 mo 26 d.

Q2. Solve the following (subtraction of time):

(a) 16 hr 49 min - 3 hr 53 min

16 hr 49 min - 3 hr 53 min: borrow 1 hr: 15 hr 109 min - 3 hr 53 min = 12 hr 56 min.

(b) 44 min 44 sec - 36 min 16 sec

44 min 44 sec - 36 min 16 sec = 8 min 28 sec.

(c) 8 weeks 1 day - 2 weeks 3 days

8 wk 1 d - 2 wk 3 d: borrow 1 week: 7 wk 8 d - 2 wk 3 d = 5 wk 5 d.

(d) 17 months - 10 months 12 days

17 months - 10 months 12 days: borrow 1 month (30 days): 16 mo 30 d - 10 mo 12 d = 6 mo 18 d.

(e) 40 months 28 days - 38 months 17 days

40 mo 28 d - 38 mo 17 d = 2 mo 11 d.

Q3. A train takes 5 hours 56 minutes to travel from Multan to Lahore and 6 hours 22 minutes to travel from Lahore to Rawalpindi. How much time does it take to travel from Multan to Rawalpindi?

5 hr 56 min + 6 hr 22 min = 11 hr 78 min = 12 hr 18 min.

The train takes 12 hr 18 min.

Q4. To complete one Science project, Hammad takes 2 weeks and 5 days and to complete another project he takes 1 week and 6 days. Which project takes more time and how much?

First project: 2 wk 5 d = 19 days. Second project: 1 wk 6 d = 13 days.

The first project takes more time, by 19 - 13 = 6 days.

Q5. Kamal's age is 10 years 5 months old and his friend's age is 11 years and 8 months. What is the difference between their ages in months?

Kamal: 10 yr 5 mo = 125 months. Friend: 11 yr 8 mo = 140 months.

Difference = 140 - 125 = 15 months.

Q6. Umer takes 3 hours 12 minutes to complete Maths homework sums and 1 hour 50 minutes to complete English homework. (a) How much time does he take to complete both tasks in minutes? (b) In which subject does he spend more time and how much more?

(a) Maths: 3 hr 12 min = 192 min. English: 1 hr 50 min = 110 min. Both = 192 + 110 = 302 min.

(b) Maths takes more time: 192 - 110 = 82 min more.

## Review Exercise - Unit 5 (book pages 105-106)

Q1. (a) There are ___ metres in 2 kilometres.

(i) 500

(ii) 1000

(iii) 200

✓ (iv) 2000

(b) To measure hours, minutes and seconds are used for ___.

✓ (i) time

(ii) distance

(iii) area

(iv) length

(c) There are ___ months in a year.

(i) 6

✓ (ii) 12

(iii) 9

(iv) 5

(d) There are ___ days in 10 months.

✓ (i) 300

(ii) 30

(iii) 15

(iv) 45

(e) There are ___ minutes in 5 hours.

(i) 60



(ii) 300

(iii) 200

(iv) 50

(f) There are ___ days in 7 weeks.



(i) 49

(ii) 42

(iii) 7

(iv) 14

Q2. Convert the following:

(a) 52 km into m

52 km = 52 000 m.

(b) 21 km 103 m into m

21 km 103 m = 21 103 m.

(c) 1 050 m into km

1 050 m = 1 km 50 m.

(d) 6 000 m into km

6 000 m = 6 km.

(e) 198 m into cm

198 m = 19 800 cm.

(f) 500 m 66 cm into cm

500 m 66 cm = 50 066 cm.

(g) 640 cm into m

640 cm = 6 m 40 cm.

(h) 98 cm into mm

98 cm = 980 mm.

Q3. Convert the following:

(a) 22 hr 66 min into min

22 hr 66 min = 22 x 60 + 66 = 1 320 + 66 = 1 386 min.

(b) 360 min into hr

360 min = 6 hr.

(c) 580 min into hr and min

580 min = 9 hr 40 min.

(d) 64 min into sec

64 min = 3 840 sec.

(e) 795 sec into min and sec

795 sec = 13 min 15 sec.

(f) 198 sec into min

198 sec = 3 min 18 sec.

Q4. Convert the following:

(a) 78 years into months

78 years = 936 months.

(b) 14 years 6 months into months

14 yr 6 mo = 174 months.

(c) 26 months into years and months

26 months = 2 yr 2 mo.

(d) 9 weeks 2 days into days

9 wk 2 d = 65 days.

(e) 35 days into weeks

35 days = 5 weeks.

(f) 420 days into months

420 days = 14 months.

Q5. Solve the following:

(a) 6 hr 52 min + 9 hr 12 min

6 hr 52 min + 9 hr 12 min = 15 hr 64 min = 16 hr 4 min.

(b) 46 min 46 sec + 11 min 10 sec

46 min 46 sec + 11 min 10 sec = 57 min 56 sec.

(c) 66 years 9 months + 22 years 7 months

66 yr 9 mo + 22 yr 7 mo = 88 yr 16 mo = 89 yr 4 mo.

(d) 54 min 20 sec - 42 min 55 sec

54 min 20 sec - 42 min 55 sec: borrow 1 min: 53 min 80 sec - 42 min 55 sec =

11 min 25 sec.

(e) 6 weeks 5 days + 11 weeks 5 days

6 wk 5 d + 11 wk 5 d = 17 wk 10 d = 18 wk 3 d.

(f) 49 months 19 days + 55 months

49 mo 19 d + 55 mo = 104 mo 19 d.

(g) 33 months 15 days - 22 months 18 days

33 mo 15 d - 22 mo 18 d: borrow 1 month (30 days): 32 mo 45 d - 22 mo 18 d = 10 mo 27 d.

(h) 20 weeks 5 days - 15 weeks 6 days

20 wk 5 d - 15 wk 6 d: borrow 1 week: 19 wk 12 d - 15 wk 6 d = 4 wk 6 d.

Q6. The lengths of two ropes are 15 m 13 cm and 12 m 42 cm respectively. (a) What is the total length of the two ropes? (b) What is the difference between their lengths?

(a) 15 m 13 cm + 12 m 42 cm = 27 m 55 cm.

(b) 15 m 13 cm - 12 m 42 cm: borrow 1 m: 14 m 113 cm - 12 m 42 cm = 2 m 71 cm.

Q7. To stitch one shirt, a tailor takes 3 hours 45 minutes and to stitch the other shirt, he takes 2 hours and 21 minutes. (a) For which shirt does he take more time and how much? (b) How much time does he take to stitch both the shirts altogether?

(a) The first shirt takes more time: 3 hr 45 min - 2 hr 21 min = 1 hr 24 min more.

(b) Both shirts: 3 hr 45 min + 2 hr 21 min = 5 hr 66 min = 6 hr 6 min.

## Unit 6: Unitary Method

### Exercise 1 - Unitary Method (book pages 107-112)

Worked example (book page 107): The price of 5 kg vegetables is Rs 400. Find the price of 13 kg vegetables.

Price of 1 kg vegetables = 400 / 5 = Rs 80.

Price of 13 kg vegetables = 80 x 13 = Rs 1 040.

Worked example (book page 108): For a science experiment, the students were divided into 8 groups and each group was given one magnifying glass. If the price of 1 magnifying glass is Rs 245, find the cost of 8 such glasses.

Price of 8 glasses = 245 x 8 = Rs 1 960.

Worked example (book page 108): If the price of a pencil is Rs 10, what will be the price of 15 pencils?

Price of 15 pencils = 10 x 15 = Rs 150.

Worked example (book page 108): The cost of 20 chairs is Rs 7 240. How can we find the cost of 1 such chair?

Price of 1 chair = 7 240 / 20 = Rs 362.

Worked example (book page 109): In a mango orchard, there are 576 trees in 32 identical rows. How many trees will be there in 1 row?

Trees in 1 row = 576 / 32 = 18 trees.

Worked example (book page 110): Father bought 12 tubelights for the home which cost Rs 6 900 altogether. What will be the price of 7 such tubelights?

Price of 1 tubelight = 6 900 / 12 = Rs 575.

Price of 7 tubelights = 575 x 7 = Rs 4 025.

Worked example (book page 110): There are 2 940 pages in 30 notebooks. How many pages will be there in 14 such notebooks?

Pages in 1 notebook = 2 940 / 30 = 98.

Pages in 14 notebooks = 98 x 14 = 1 372.

Q1. The price of 1 geometry box is Rs 76. Find out the price of 26 such geometry boxes.

Price of 26 geometry boxes = 76 x 26 = Rs 1 976.

Q2. If the price of one ice cream is Rs 55, find out the price of 17 such ice-creams.

Price of 17 ice-creams = 55 x 17 = Rs 935.

Q3. If the price of a popcorn pack is Rs 37, find out the price of 71 such popcorn packs.

Price of 71 popcorn packs = 37 x 71 = Rs 2 627.

Q4. A bag weighs 21 kg. What will be the weight of 15 such bags?

Weight of 15 bags = 21 x 15 = 315 kg.

Q5. The capacity of 15 identical buses is 555 passengers. What will be the capacity of 1 bus?

Capacity of 1 bus = 555 / 15 = 37 passengers.

Q6. The price of 26 kg sugar is Rs 1 170. What will be the price of 1 kg of sugar?

Price of 1 kg sugar = 1 170 / 26 = Rs 45.

Q7. If the rent of a house for 8 months is Rs 145 680, find out the rent of the house for 1 month.

Rent for 1 month = 145 680 / 8 = Rs 18 210.

Q8. The cost of 35 registers is Rs 5 075. Find out the cost of 1 register.

Cost of 1 register = 5 075 / 35 = Rs 145.

## Exercise 2 - Unitary Method (continued, book page 111)

Q1. There are 4 900 markers in 50 boxes. How many markers will be there in 75 such boxes?

Markers in 1 box = 4 900 / 50 = 98.

Markers in 75 boxes = 98 x 75 = 7 350.

Q2. The price of 10 mobile phones is Rs 348 290. What will be the price of 29 mobile phones of the same model?

Price of 1 mobile phone = 348 290 / 10 = Rs 34 829.

Price of 29 mobile phones = 34 829 x 29 = Rs 1 010 041.

Q3. 40 books weigh 88 kg. What is the weight of 6 such books?

Weight of 1 book = 88 / 40 = 2.2 kg.

Weight of 6 books = 2.2 x 6 = 13.2 kg.

Q4. A train travels 6 136 km in 52 hours. How much distance will it cover at the same speed in 45 hours?

Distance in 1 hour = 6 136 / 52 = 118 km.

Distance in 45 hours = 118 x 45 = 5 310 km.

Q5. Amir buys 3 computers for Rs 838 155. What will be the cost of 52 such computers?

Cost of 1 computer = 838 155 / 3 = Rs 279 385.

Cost of 52 computers = 279 385 x 52 = Rs 14 528 020.

Q6. If the fare for 34 km is Rs 1 190, what will be the fare for 48 km?

Fare for 1 km = 1 190 / 34 = Rs 35.

Fare for 48 km = 35 x 48 = Rs 1 680.

Q7. 19 pairs of shoes cost Rs 23 750. What will be the price of 65 such pairs?

Price of 1 pair = 23 750 / 19 = Rs 1 250.

Price of 65 pairs = 1 250 x 65 = Rs 81 250.

Review Exercise - Unit 6 (book pages 111-112)

Q1. (a) The price of a book is Rs 250, the price of 5 books will be Rs ___.

(i) 50

(ii) 1000

✓ (iii) 1250

(iv) 2500

(b) The price of 11 carpets is Rs 35 805, the price of 1 carpet will be Rs ___.

(i) 3855

(ii) 3055

(iii) 2355

✓ (iv) 3255

(c) The price of a book is Rs 555. The price of 10 books will be Rs ___.

(i) 2550

(ii) 5050

(iii) 3250

✓ (iv) 5550

(d) The price of 6 oranges is Rs 48. The price of 72 oranges will be Rs ___.

(i) 567

(ii) 96

(iii) 112

✓ (iv) 576

(e) The price of 3 chairs is Rs 645. The price of 16 chairs will be Rs ___.

(i) 3040

(ii) 3004

✓ (iii) 3440

(iv) 3404

Q2. The price of a storybook is Rs 440. Find out the price of 15 such storybooks.

Price of 15 storybooks = 440 x 15 = Rs 6 600.

Q3. If the price of 16 school bags is Rs 24 000, find out the price of 1 such bag.

Price of 1 bag = 24 000 / 16 = Rs 1 500.

Q4. 425 pearls were used to make a necklace. Find: (a) how many pearls will be used to make 26 necklaces? (b) how many pearls will be used to make 100 necklaces?

(a) 425 x 26 = 11 050 pearls.

(b) 425 x 100 = 42 500 pearls.

Q5. If the price of 5 washing machines is Rs 92 465, then find: (a) the price of 24 such washing machines (b) the price of 10 such washing machines.

Price of 1 washing machine = 92 465 / 5 = Rs 18 493.

(a) Price of 24 machines = 18 493 x 24 = Rs 443 832.

(b) Price of 10 machines = 18 493 x 10 = Rs 184 930.

## Unit 7: Geometry

### Exercise 1 - Angles (book pages 114-120)

Key facts: What is a straight angle? What is a reflex angle? What is the standard unit for measuring angles?

A straight angle measures 180 degrees.

A reflex angle is greater than 180 degrees but smaller than a complete rotation of 360 degrees.

The standard unit for measuring angles is the degree, written as degrees. One degree is 1/360 of a complete revolution.

Try Yourself (book page 115): How many 30 degree angles are there in a right angle?

A right angle = 90 degrees. 90 / 30 = 3.

There are three 30 degree angles in a right angle.

Try Yourself (book page 116): (1) How many right angles are there in a straight angle? (2) How many straight angles are there in a complete rotation?

(1) A straight angle = 180 degrees, a right angle = 90 degrees. 180 / 90 = 2 right angles.

(2) A complete rotation = 360 degrees. 360 / 180 = 2 straight angles.

Worked construction (book page 116): Construct an angle ABC = 285 degrees by using a protractor.

A reflex angle of 285 degrees: first subtract from 360 degrees: 360 - 285 = 75 degrees.

Draw a ray BA, place the protractor upside down on ray BA, mark a point C at 75 degrees, join B to C.

Angle ABC = 285 degrees is the required reflex angle.

Pairs of angles: What are adjacent angles? Give an example from the book.

Adjacent angles are two angles that have a common vertex and a common side, and lie on opposite sides of the common side.

Example: Angle AOB is adjacent to angle BOC. Their common arm is OB and the common vertex is O.

Pairs of angles: What are complementary angles? Give an example from the book.

Two angles are called complementary angles if their sum is 90 degrees. One angle is the complement of the other. They may be adjacent or non-adjacent.

Example: Angle AOD + Angle DOB = 45 degrees + 45 degrees = 90 degrees.

Try Yourself (book page 118): Can two right angles be the complementary angles of each other? Explain your answer.

No. Two right angles add to 90 + 90 = 180 degrees, not 90 degrees. So they cannot be complementary angles of each other.

Pairs of angles: What are supplementary angles? Give an example from the book.

Two angles are called supplementary angles if their sum is 180 degrees. They may be adjacent or non-adjacent.

Example: Angle AOD + Angle DOC = 100 degrees + 80 degrees = 180 degrees.

Try Yourself (book page 119): Can two straight angles be the supplementary angles of each other? Explain your answer.

No. Two straight angles add to 180 + 180 = 360 degrees, not 180 degrees. So they cannot be supplementary angles of each other.

Q1. Draw a right angle by using cardboard and compare it with the given angles and indicate their types. (Less than 90 degrees = acute angle, equal to 90 degrees = right angle, greater than 90 degrees but less than 180 degrees = obtuse angle, equal to 180 degrees = straight angle, greater than 180 degrees = reflex angle.)

Q2. Use the protractor to draw the given angles: 35 degrees, 45 degrees, 240 degrees, 180 degrees, 90 degrees, 60 degrees, 300 degrees, 155 degrees.

Q3. Identify adjacent angles from the given angles. (Adjacent angles share a common vertex and a common side.)

Q4. Find the supplements of the following:

Supplement of 98 degrees

180 - 98 = 82 degrees.

Supplement of 180 degrees

180 - 180 = 0 degrees.

Supplement of 115 degrees

180 - 115 = 65 degrees.

Supplement of 3 degrees

180 - 3 = 177 degrees.

Supplement of 78 degrees

180 - 78 = 102 degrees.

Supplement of 135 degrees

180 - 135 = 45 degrees.

Supplement of 90 degrees

180 - 90 = 90 degrees.

Q5. Find the complements of the following:

Complement of 12 degrees

90 - 12 = 78 degrees.

Complement of 88 degrees

90 - 88 = 2 degrees.

Complement of 90 degrees

90 - 90 = 0 degrees.

Complement of 10 degrees

90 - 10 = 80 degrees.

Complement of 62 degrees

90 - 62 = 28 degrees.

Complement of 40 degrees

90 - 40 = 50 degrees.

Complement of 70 degrees

90 - 70 = 20 degrees.

## Exercise 2 - Triangles (book pages 121-127)

Types of triangles with respect to their sides: define equilateral, isosceles and scalene triangles.

Equilateral triangle: all three sides are equal in length. All angles of an equilateral triangle are equal to 60 degrees.

Isosceles triangle: two sides are equal in length. Two base angles of an isosceles triangle are equal in measure.

Scalene triangle: all sides are of different lengths. All angles in a scalene triangle are different in measure.

Try Yourself (book page 122): Is every equilateral triangle an isosceles triangle? Explain your answer.

Yes. An isosceles triangle is a triangle in which two sides are equal. An equilateral triangle has all three sides equal, so it certainly has two equal sides. Hence every equilateral triangle is also an isosceles triangle.

Types of triangles with respect to their angles: define acute angled, right angled and obtuse angled triangles, with the book's examples.

Acute angled triangle: all angles are acute (less than 90 degrees). Example: triangle FGH with angles 40, 60 and 80 degrees.

Right angled triangle: one angle is a right angle (90 degrees). Example: triangle DEF with angle E = 90 degrees.

Obtuse angled triangle: one angle is obtuse (greater than 90 degrees). Example: triangle LKM with angle K = 120 degrees.

Construction (book page 124): Construct a triangle EFG in which EF = 6 cm, angle E = 35 degrees and angle F = 70 degrees. Also find the measurement of the remaining angle.

Draw EF = 6 cm. Draw an angle of 35 degrees at E and an angle of 70 degrees at F. The two rays intersect at G.

Angle G = 180 - (35 + 70) = 180 - 105 = 75 degrees.

Measure EG and FG with a ruler.

Construction (book page 125): Construct a triangle ABC in which AB = 4 cm, angle B = 30 degrees and BC = 3 cm. Measure the remaining sides and angles with a ruler and protractor.

Q1. Identify these triangles with respect to their sides. (All sides equal = equilateral, two sides equal = isosceles, all sides different = scalene.)

Q2. Identify these triangles with respect to their angles. (All angles less than 90 degrees = acute angled, one angle 90 degrees = right angled, one angle greater than 90 degrees = obtuse angled.)

Q3. Draw a triangle ABC in which angle A = 75 degrees, AB = 4 cm and angle B = 55 degrees. What is angle C?

Draw AB = 4 cm, draw 75 degrees at A and 55 degrees at B. The rays meet at C.

Angle C = 180 - (75 + 55) = 180 - 130 = 50 degrees.

Q4. Draw a triangle LMN in which angle L = 60 degrees, LM = 5.5 cm and angle M = 30 degrees. What is angle N?

Draw LM = 5.5 cm, draw 60 degrees at L and 30 degrees at M. The rays meet at N.

Angle N = 180 - (60 + 30) = 90 degrees.

Q5. Draw a triangle JKL in which KL = 4 cm, JK = 3.2 cm and angle K = 30 degrees.

Q6. Draw a triangle XYZ in which YZ = 7.3 cm, XY = 3.5 cm and angle Y = 40 degrees.

## Exercise 3 - Quadrilaterals (book pages 127-133)

**What is a quadrilateral? Describe the six kinds taught in the book: square, rectangle, parallelogram, rhombus, trapezium and kite.**

A quadrilateral is a shape with four sides and four angles.

Square: four equal sides and four right angles.

Rectangle: two pairs of parallel sides, opposite sides equal, all angles 90 degrees.

Parallelogram: two pairs of parallel sides, opposite angles equal, none of the angles a right angle.

Rhombus: all sides equal and parallel, opposite angles equal but not right angles.

Trapezium: only one pair of parallel sides.

Kite: two equal pairs of adjacent sides.

**Try It (book page 131): Is every square a rectangle? Is every rectangle a square? Is every rhombus a parallelogram? Is every parallelogram a rhombus? Explain your answers.**

Every square is a rectangle: yes, because a square has opposite sides equal and all angles 90 degrees, which is the definition of a rectangle.

Every rectangle is a square: no, because a rectangle need not have all four sides equal.

Every rhombus is a parallelogram: yes, because a rhombus has two pairs of parallel sides.

Every parallelogram is a rhombus: no, because a parallelogram need not have all sides equal.

Q1. Label the following quadrilaterals and describe at least two properties of each (in terms of sides and angles).

Q2. Draw squares according to the given lengths by using protractor and ruler: 3.4 cm, 5 cm, 7.2 cm, 4 cm, 6.3 cm, 5.5 cm, 8.9 cm, 3.3 cm.

Q3. Draw rectangles by using protractor and ruler according to the given lengths and widths: (7 cm, 5 cm), (8 cm, 4.4 cm), (6.6 cm, 3.4 cm), (8.5 cm, 5 cm), (6 cm, 3.2 cm), (9 cm, 5.2 cm).

## Exercise 4 - Symmetry (book pages 133-136)

**What is reflective symmetry? How many lines of symmetry do a square and a rectangle have?**

Reflective symmetry: a figure has reflective symmetry about a line if folding it along that line makes one half fit exactly onto the other half. The line is called the line of symmetry.

A square has four lines of symmetry. A rectangle has two lines of symmetry.

**What is rotational symmetry? What is the order of rotational symmetry in the book's examples?**

Rotational symmetry: a figure has rotational symmetry if, when rotated about its centre, it looks exactly the same more than once in a complete revolution. The number of times it looks the same is the order of rotational symmetry.

The book's shape looks the same 4 times in a complete revolution (at 90, 180, 270 and 360 degrees), so its order of rotational symmetry is 4.

Another figure looks the same twice, so its order is 2. A figure that looks the same only once has no rotational symmetry.

Q1. Encircle the figures which have reflective symmetry. Also draw their lines of symmetry.

Q2. Encircle the figures having rotational symmetry. Also write the order of their rotation and mark their centre of rotation.

## Exercise 5 - Nets of 3-D Shapes (book pages 137-140)

**What is a net? Describe the nets of a cube, a cuboid and a square-based pyramid.**

A net is a 2-dimensional shape which can be folded in a specific pattern to get a 3-dimensional solid. A 3-dimensional solid can have more than one possible net.

Cube: its net is made of 6 similar squares.

Cuboid: its net is made of 6 rectangles.

Square-based pyramid: its net is made of 4 triangles and 1 square.

Q1. Write the names of these figures. Also write the number and name of faces.

Q2. Identify the nets of a cube and colour them.

Q3. Observe these nets and identify the 3-D solid they are made of.

Review Exercise - Unit 7 (book pages 142-144)

Q1. (a) Which of these is a reflex angle?

(i) 375 degrees

✓ (ii) 215 degrees

(iii) 180 degrees

(iv) 90 degrees

(b) The supplement of 20 degrees is:

✓ (i) 160 degrees

(ii) 40 degrees

(iii) 70 degrees

(iv) 80 degrees

(c) Sum of two right angles is equal to:

(i) Reflex angle

✓ (ii) Straight angle

(iii) Acute angle

(iv) Obtuse angle

(d) Two angles will be called supplementary angles if their sum is equal to:

✓ (i) 180 degrees

(ii) 90 degrees

(iii) 360 degrees

(iv) 100 degrees

Q1. (e) Which of the following shapes is not a quadrilateral?

The pentagon (five-sided figure) is not a quadrilateral. A quadrilateral has four sides.

(f) A triangle in which its ___ sides are equal, is called an isosceles triangle.

(i) 1

✓ (ii) 2

(iii) 3

(iv) 4

Q1. (g) Which of the following is not the net of a cube? (A net of a cube has 6 squares that fold into a cube; the figure that cannot fold into a cube is not its net.)

Q1. (h) The order of the rotational symmetry of the shape C is: (Count how many times the shape looks exactly the same in one complete revolution.)

Q1. (i) Which of the following is showing adjacent angles? (Adjacent angles share a common vertex and a common side.)

Q2. Draw these angles by using protractor and ruler: 35 degrees, 45 degrees, 240 degrees, 180 degrees, 90 degrees, 60 degrees, 300 degrees, 155 degrees.

Q3. Identify the adjacent angles from the given figures.

Q4. Make 5 pairs of each complementary and supplementary angles.

Complementary pairs (sum 90 degrees): (30, 60), (45, 45), (20, 70), (10, 80), (35, 55).

Supplementary pairs (sum 180 degrees): (100, 80), (120, 60), (90, 90), (150, 30), (45, 135).

Q5. How many types of triangles are there with respect to their sides and angles?

With respect to sides, there are 3 types: equilateral, isosceles and scalene.

With respect to angles, there are 3 types: acute angled, right angled and obtuse angled.

Q6. Draw a triangle IJK in which angle I = 70 degrees, IJ = 6.8 cm and angle J = 28 degrees. What is angle K?

Draw IJ = 6.8 cm, draw 70 degrees at I and 28 degrees at J. The rays meet at K.

Angle K = 180 - (70 + 28) = 82 degrees.

Q7. Draw a triangle PQR in which QR = 3.3 cm, PQ = 5.2 cm and angle PQR = 75 degrees.

Q8. Draw squares according to the given lengths with the help of protractor and ruler: 4 cm, 5.1 cm, 3.6 cm, 4.9 cm.

Q9. Draw rectangles with the help of protractor and ruler according to the given lengths and widths: (length 4 cm, width 3.4 cm), (length 5 cm, width 3 cm), (length 6.6 cm, width 4.2 cm), (length 7 cm, width 2.4 cm).

Q10. Encircle the figures which have reflective symmetry. Also draw their lines of symmetry.

Q11. Encircle the figures having rotational symmetry. Also write the order of their rotation and mark their centre of rotation.

Q12. Use cardboard to make nets of various solids. Also write the number of their faces and the shape. Then fold them and verify whether you have created the correct net or not.

## Unit 8: Perimeter and Area

### Exercise 1 - Perimeter (book pages 146-151)

Key facts: What is area? What is perimeter? What are their units?

Area: the space covered by the surface of any 2-dimensional shape.

Perimeter: the total length of the boundary of a closed region.

Units of area: square metres (m squared) and square centimetres (cm squared).

Units of perimeter: metres (m) and centimetres (cm).

Worked example (book page 147): Usman has a square shaped picture frame with length of one side 22.5 cm. Find the perimeter of the frame.

Perimeter of a square = 4 x length of one side.

= 4 x 22.5 cm = 90 cm.

So, the perimeter of the picture frame is 90 cm.

Try Yourself (book page 148): A square shaped ground is 20 metres long. If a person takes 5 rounds of the ground, how much total distance does he cover?

Perimeter of the ground = 4 x 20 m = 80 m.

Distance in 5 rounds = 5 x 80 m = 400 m.

So, the person covers a total distance of 400 m.

Worked example (book page 149): A rectangular pool is 10 metres long and 7.3 metres wide. Find the perimeter of the pool.

Perimeter of rectangle = 2 (length + width).

= 2 (10 m + 7.3 m) = 2 (17.3 m) = 34.6 m.

So, the perimeter of the pool is 34.6 m.

Worked example (book page 149): The length of a rectangular garden is 21 metres and its width is 16 metres. If the rate of fencing is Rs. 170 per metre, find the cost of fencing around it.

Perimeter of the garden = 2 (21 + 16) = 2 (37) = 74 m.

Cost of fencing = 74 x Rs. 170 = Rs. 12,580.

So, the cost of fencing is Rs. 12,580.

Try Yourself (book page 149): The perimeter of a rectangular field is 66 metres. If the length of this field is 15 metres, find the width of the field. Also find the cost of putting barbed wire around the field if the rate is Rs. 90 per metre.

Perimeter = 2 (length + width).

66 = 2 (15 + width).

66 = 30 + 2 x width.

2 x width = 66 - 30 = 36.

Width = 36 / 2 = 18 m.

Cost of barbed wire = 66 x Rs. 90 = Rs. 5,940.

Q1. Find the perimeter of the given squares.

Square of side 7 m

Perimeter = 4 x 7 m = 28 m.

Square of side 13 cm

Perimeter = 4 x 13 cm = 52 cm.

Square of side 4 m

Perimeter = 4 x 4 m = 16 m.

Q2. Find the perimeter of the given rectangles.

Rectangle 9 cm by 5 cm

Perimeter = 2 (9 + 5) = 2 (14) = 28 cm.

Rectangle 3 m by 8 m

Perimeter = 2 (3 + 8) = 2 (11) = 22 m.

Rectangle 12 m by 2 m

Perimeter = 2 (12 + 2) = 2 (14) = 28 m.

Q3. If the length of a square shaped crop field is 29 m, what will be its perimeter?

Perimeter = 4 x 29 m = 116 m.

Q4. The perimeter of a square shape is 72 cm. What will be its length?

Length = Perimeter / 4 = 72 / 4 = 18 cm.

Q5. Find the perimeter of the squares of the given lengths by using formula.

Square of side 5 cm

Perimeter = 4 x 5 = 20 cm.

Square of side 12 cm

Perimeter = 4 x 12 = 48 cm.

Square of side 6 m

Perimeter = 4 x 6 = 24 m.

Square of side 19 cm

Perimeter = 4 x 19 = 76 cm.

Square of side 26 m

Perimeter = 4 x 26 = 104 m.

Square of side 2.5 cm

Perimeter = 4 x 2.5 = 10 cm.

Square of side 9.7 m

Perimeter = 4 x 9.7 = 38.8 m.

Square of side 15 m

Perimeter = 4 x 15 = 60 m.

Square of side 16.6 cm

Perimeter = 4 x 16.6 = 66.4 cm.

Square of side 10 m

Perimeter = 4 x 10 = 40 m.

Square of side 7.1 cm

Perimeter = 4 x 7.1 = 28.4 cm.

Square of side 2.7 cm

Perimeter = 4 x 2.7 = 10.8 cm.

Q6. Find the perimeter of the rectangles of the given lengths and widths by using formula.

Rectangle length 3 cm, width 2 cm

Perimeter = 2 (3 + 2) = 10 cm.

Rectangle length 5.3 m, width 2.2 m

Perimeter = 2 (5.3 + 2.2) = 15 m.

Rectangle length 6 cm, width 4 cm

Perimeter = 2 (6 + 4) = 20 cm.

Rectangle length 9 m, width 1.2 m

Perimeter = 2 (9 + 1.2) = 20.4 m.

Rectangle length 10 m, width 5.9 m

Perimeter = 2 (10 + 5.9) = 31.8 m.

Rectangle length 15 cm, width 12 cm

Perimeter = 2 (15 + 12) = 54 cm.

Q7. Children are playing in a square shaped playground. If the length of the playground is 12 metres, find its perimeter.

Perimeter = 4 x 12 m = 48 m.

Q8. Harris wants to find out the perimeter of the square shaped notice board in his classroom. If the length of one side of the notice board is 2.5 metres, find the perimeter of the notice board.

Perimeter = 4 x 2.5 m = 10 m.

Q9. If a rectangular room is 10.8 metres long and 8.8 metres wide. Find the perimeter of the room.

Perimeter = 2 (10.8 + 8.8) = 2 (19.6) = 39.2 m.

Q10. Nadia has a rectangular frame. The frame is 12 cm long and 8 cm wide. Nadia wants to put a ribbon around the frame. (a) Find the required length of the ribbon. (b) What will be the total cost of ribbon if 1 metre of it costs Rs. 5?

(a) Perimeter of the frame = 2 (12 + 8) = 2 (20) = 40 cm. So, the required length of ribbon is 40 cm.

(b) 40 cm = 0.4 m. Cost = 0.4 x Rs. 5 = Rs. 2.

Q11. A building is 128 metres long and 96.5 metres wide. (a) Find out its perimeter. (b) Find the total cost for the construction of boundary wall around this building if the rate of construction of wall is Rs. 470 per metre.

(a) Perimeter = 2 (128 + 96.5) = 2 (224.5) = 449 m.

(b) Cost = 449 x Rs. 470 = Rs. 211,030.

## Exercise 2 - Area (book pages 151-155)

Worked example (book page 151): A square mirror has a length of 42 cm. What is its area?

Area of a square = length x length.

= 42 cm x 42 cm = 1,764 cm squared.

So, the area of the mirror is 1,764 cm squared.

Worked example (book page 152): The playground of my school has a rectangular flower bed with length 7.5 metres and width 4 metres. Find its area.

Area of rectangle = length x width.

= 7.5 m x 4 m = 30 m squared.

So, the area of the flower bed is 30 m squared.

Worked example (book page 153): A rectangular hospital has a perimeter equal to 294 metres. If its length is 85 metres, find its area.

Perimeter = 2 (length + width).

294 = 2 (85 + width).

294 = 170 + 2 x width.

2 x width = 294 - 170 = 124.

Width = 124 / 2 = 62 m.

Area = 85 m x 62 m = 5,270 m squared.

So, the area of the hospital is 5,270 m squared.

Q1. Find the area of the given squares.

Square of side 2.5 m

Area = 2.5 x 2.5 = 6.25 m squared.

Square of side 22 cm

Area = 22 x 22 = 484 cm squared.

Square of side 42.7 m

Area = 42.7 x 42.7 = 1823.29 m squared.

Q2. Find the area of the given rectangles.

Rectangle 4.5 cm by 3 cm

Area = 4.5 x 3 = 13.5 cm squared.

Rectangle 2 m by 6 m

Area = 2 x 6 = 12 m squared.

Rectangle 10 m by 1 m

Area = 10 x 1 = 10 m squared.

Q3. The area of a rectangle is 96 m squared. If its width is 3 m, then find its length.

Length = Area / width = 96 / 3 = 32 m.

Q4. Find the area of the squares of the given lengths by using formula.

Square of side 4.5 m

Area = 4.5 x 4.5 = 20.25 m squared.

Square of side 9.3 cm

Area = 9.3 x 9.3 = 86.49 cm squared.

Square of side 8.8 m

Area = 8.8 x 8.8 = 77.44 m squared.

Square of side 15 cm

Area = 15 x 15 = 225 cm squared.

Square of side 13 cm

Area = 13 x 13 = 169 cm squared.

Square of side 3 m

Area = 3 x 3 = 9 m squared.

Square of side 6 m

Area = 6 x 6 = 36 m squared.

Square of side 2.9 m

Area = 2.9 x 2.9 = 8.41 m squared.

Square of side 5 cm

Area = 5 x 5 = 25 cm squared.

Square of side 9.2 m

Area = 9.2 x 9.2 = 84.64 m squared.

Square of side 14 m

Area = 14 x 14 = 196 m squared.

Square of side 1.1 cm

Area = 1.1 x 1.1 = 1.21 cm squared.

Q5. Find the area of the rectangles of the given lengths and widths by using formula.

Rectangle length 5 cm, width 1.9 cm

Area = 5 x 1.9 = 9.5 cm squared.

Rectangle length 4 m, width 3 m

Area = 4 x 3 = 12 m squared.

Rectangle length 6 cm, width 4 cm

Area = 6 x 4 = 24 cm squared.

Rectangle length 7 cm, width 5 cm

Area = 7 x 5 = 35 cm squared.

Rectangle length 10.5 cm, width 9 cm

Area = 10.5 x 9 = 94.5 cm squared.

Rectangle length 20 m, width 17 m

Area = 20 x 17 = 340 m squared.

Q6. A rectangular shaped ground has a length of 122 m and width 108 m. Find the area of the ground.

Area = 122 x 108 = 13,176 m squared.

Q7. The area of a school's main gate is 19.55 m squared. (a) If the width of the gate is 2.3 m, then find its length. (b) Find the cost of painting the gate if the rate of painting is Rs. 275 per m squared.

(a) Length = Area / width = 19.55 / 2.3 = 8.5 m.

(b) Cost = 19.55 x Rs. 275 = Rs. 5,376.25.

Q8. Area of a Masjid is 27,540 m squared and its length is 255 m. Find: (a) The perimeter of the Masjid. (b) The cost of carpeting the Masjid, if the rate of carpeting is Rs. 275 per m squared.

(a) Width = Area / length = 27,540 / 255 = 108 m.

Perimeter = 2 (255 + 108) = 2 (363) = 726 m.

(b) Cost = 27,540 x Rs. 275 = Rs. 7,573,500.

Q9. Find the area of the blue part. (The figure is a rectangle 22 cm by 10 cm, with a white rectangle 6 cm by 2 cm and a dark square 4 cm by 4 cm cut out of it.)

Area of the big rectangle = 22 x 10 = 220 cm squared.

Area of the white rectangle = 6 x 2 = 12 cm squared.

Area of the dark square = 4 x 4 = 16 cm squared.

Area of the blue part = 220 - 12 - 16 = 192 cm squared.

## Review Exercise - Unit 8 (book pages 156-157)

Q1. (a) If the length of a rectangle is 4 cm and width is 3.4 cm, then its perimeter will be equal to:

(i) 11.4 cm

(ii) 7.4 cm

✓ (iii) 14.8 cm

(iv) 40.8 cm

(b) Formula to find the perimeter of the square is:

(i) 4 + length

(ii) 4 - length

✓ (iii) 4 x length

(iv) 2 x length

(c) The formula to find the perimeter of the rectangle is:

✓ (i) 2 (length + width)

(ii) 2 x length + width

(iii) length + width

(iv) length x width

(d) Area of a rectangle is 45 m squared. If its length is 15 m, then its width is:

(i) 6 m

✓ (ii) 3 m

(iii) 5 m

(iv) 15 m

(e) The formula to find the area of the square is:

✓ (i) length x length

(ii) 2 (length + width)

(iii) length + length

(iv) 4 x length

(f) The formula to find the area of the rectangle is:

✓ (i) length x width

(ii) 2 (length + width)

(iii) length + length

(iv) 4 x length

(g) If the perimeter of the rectangle is 34 cm and we increase its length by 2 cm, then there will be difference of ___ cm in its perimeter.

(i) 2

✓ (ii) 4

(iii) 8

(iv) 6

(h) If the length of one side of the square is 14 cm, then its perimeter will be:

(i) 14 cm

✓ (ii) 56 cm

(iii) 256 cm

(iv) 28 cm

Q2. Find the perimeter and area of the squares of the given lengths by using formula.

Square of side 8.2 cm

Perimeter = 4 x 8.2 = 32.8 cm.

Area = 8.2 x 8.2 = 67.24 cm squared.

Square of side 2.6 m

Perimeter = 4 x 2.6 = 10.4 m.

Area = 2.6 x 2.6 = 6.76 m squared.

Square of side 12.8 m

Perimeter = 4 x 12.8 = 51.2 m.

Area = 12.8 x 12.8 = 163.84 m squared.

Square of side 7.9 cm

Perimeter = 4 x 7.9 = 31.6 cm.

Area = 7.9 x 7.9 = 62.41 cm squared.

Square of side 16 cm

Perimeter = 4 x 16 = 64 cm.

Area = 16 x 16 = 256 cm squared.

Square of side 4.3 m

Perimeter = 4 x 4.3 = 17.2 m.

Area = 4.3 x 4.3 = 18.49 m squared.

Square of side 5.7 m

Perimeter = 4 x 5.7 = 22.8 m.

Area = 5.7 x 5.7 = 32.49 m squared.

Square of side 11 cm

Perimeter = 4 x 11 = 44 cm.

Area = 11 x 11 = 121 cm squared.

Q3. Find the perimeter and area of the rectangles of the given lengths and widths by using formula.

Rectangle length 6 cm, width 3.4 cm

Perimeter = 2 (6 + 3.4) = 18.8 cm.

Area = 6 x 3.4 = 20.4 cm squared.

Rectangle length 1.2 m, width 0.3 m

Perimeter = 2 (1.2 + 0.3) = 3 m.

Area = 1.2 x 0.3 = 0.36 m squared.

Rectangle length 10 cm, width 13 cm

Perimeter = 2 (10 + 13) = 46 cm.

Area = 10 x 13 = 130 cm squared.

Rectangle length 17 cm, width 8.5 cm

Perimeter = 2 (17 + 8.5) = 51 cm.

Area = 17 x 8.5 = 144.5 cm squared.

Q4. The perimeter of a book is 100 cm. If its width is 22 cm, find its length.

Perimeter = 2 (length + width).

100 = 2 (length + 22).

length + 22 = 100 / 2 = 50.

length = 50 - 22 = 28 cm.

Q5. Laiba wants tiling for the floor of her kitchen. If the length of the kitchen is 3 metres and width is 2.5 metres, then find the area of the kitchen.

Area = 3 x 2.5 = 7.5 m squared.

Q6. The length of the fence around a square shaped garden is 24 m. Find the length of the garden.

Length = Perimeter / 4 = 24 / 4 = 6 m.

Q7. Daniya took a 42 cm long ribbon and made a rectangle with it. If the length of the rectangle is 15 cm, then find its width.

Perimeter of the rectangle = 42 cm.

2 (length + width) = 42.

length + width = 42 / 2 = 21.

width = 21 - 15 = 6 cm.

Q8. Find the area of the carrom board whose perimeter is 40 cm.

A carrom board is square shaped.

Side = Perimeter / 4 = 40 / 4 = 10 cm.

Area = 10 x 10 = 100 cm squared.

# Unit 9: Data Handling

## Introduction - Reading a Bar Graph (book page 158)

Umair drew a graph of monthly average temperature for the first 5 months of 2020. The table is: January 18 degrees C, February 19 degrees C, March 21 degrees C, April 24 degrees C, May 28 degrees C. Is this graph correct? If not, then identify the mistake and correct it.

No, the graph is not fully correct. The bars for January and May are drawn too high: January's bar reaches about 19 degrees C instead of 18 degrees C, and May's bar reaches about 29 degrees C instead of 28 degrees C.

The corrected bar heights are: January 18 degrees C, February 19 degrees C, March 21 degrees C, April 24 degrees C, May 28 degrees C.

## Exercise 1 - Average (book pages 159-161)

Key fact: How do we find the average of given quantities?

To find the average of the given quantities, we find the sum of the quantities and divide this sum by the number of quantities.

Average = Sum of quantities / Number of quantities.

Worked example (book page 159): I collected 16 seashells, my brother collected 12 seashells and my sister collected 11 seashells. How can we divide these seashells equally among us?

Total number of seashells = 16 + 12 + 11 = 39.

Number of persons = 3.

Number of seashells each person will get = 39 / 3 = 13.

Every person will get 13 seashells. So, 13 is the average of 16, 12 and 11.

Try Yourself (book page 159): Find the average of 13, 34, 16 and 28.

Sum = 13 + 34 + 16 + 28 = 91.

Number of quantities = 4.

Average = 91 / 4 = 22.75.

Worked example (book page 160): The length of 5 ropes is 2.4 m, 0.9 m, 1.6 m, 4.4 m and 3 m respectively. Find the average length of these ropes.

Sum of the lengths = 2.4 + 0.9 + 1.6 + 4.4 + 3 = 12.3 m.

Number of ropes = 5.

Average = 12.3 / 5 = 2.46 m.

So, the average length of the rope is 2.46 m.

Worked example (book page 160): In the annual exams, Ali got an average of 78 marks in 7 subjects. Find the total marks he got.

Total marks obtained = Average marks obtained x Number of subjects.

= 78 x 7 = 546.

So, Ali got 546 marks in 7 subjects.

Try Yourself (book page 160): Find the average of 11, 23, 37, 55 and 82.

Sum = 11 + 23 + 37 + 55 + 82 = 208.

Number of quantities = 5.

Average = 208 / 5 = 41.6.

Try Yourself (book page 160): If the sum of quantities is 1,840 and their average is 115, then find the number of quantities.

Number of quantities = Sum of quantities / Average.

= 1,840 / 115 = 16.

Try Yourself (book page 160): If the average of 40 quantities is 215, then find the sum of the quantities.

Sum of quantities = Average x Number of quantities.

= 215 x 40 = 8,600.

Try Yourself Challenge (book page 161): Find the number of quantities, if their sum is 5,200 and their average is 650.

Number of quantities = Sum / Average = 5,200 / 650 = 8.

Try Yourself Challenge (book page 161): Yousaf took 4 mathematics tests in 4 weeks. The average marks he got in these 4 tests is 62. If he scored 80 marks in the test taken on the fifth week, find out the new average of his total marks for the five weeks overall.

Total marks in 4 tests = 62 x 4 = 248.

Total marks in 5 tests = 248 + 80 = 328.

New average = 328 / 5 = 65.6.

Q1. Find the average of the following:

(a) 4, 13, 25, 32, 42, 52

Sum = 4 + 13 + 25 + 32 + 42 + 52 = 168.

Number of quantities = 6.

Average = 168 / 6 = 28.

(b) 12 kg, 16 kg, 26 kg, 42 kg

Sum = 12 + 16 + 26 + 42 = 96 kg.

Number of quantities = 4.

Average = 96 / 4 = 24 kg.

(c) 10 cm, 13 cm, 17 cm, 16 cm, 19 cm

Sum = 10 + 13 + 17 + 16 + 19 = 75 cm.

Number of quantities = 5.

Average = 75 / 5 = 15 cm.

(d) 5, 15, 30, 25, 40 (litres)

Sum = 5 + 15 + 30 + 25 + 40 = 115.

Number of quantities = 5.

Average = 115 / 5 = 23.

Q2. Munir recites 9, 11, 12, 10 and 8 Ayah of the Holy Quran in 5 days respectively. Find the average number of Ayah she recites in one day.

Sum = 9 + 11 + 12 + 10 + 8 = 50.

Number of days = 5.

Average = 50 / 5 = 10 Ayah per day.

Q3. Saad has 16, Amna has 20, Sara has 15 and Ahmad has 9 pencils. Find the average of pencils.

Sum = 16 + 20 + 15 + 9 = 60.

Number of children = 4.

Average = 60 / 4 = 15 pencils.

Q4. The average number of students in 18 schools situated in a city is 1,150. Find the total number of students in these schools altogether.

Total students = 1,150 x 18 = 20,700.

Q5. On an average, Maryam baked 27 cakes for a bakery in 11 months. What is the total number of cakes that she baked?

Total cakes = 27 x 11 = 297.

Q6. A factory hired a total of 1,240 labourers in 4 years. Find the average number of labourers hired in a year.

Average = 1,240 / 4 = 310 labourers per year.

Q7. The runs scored by the students of a class in a cricket match are given below: Aimen 21, Haniya 52, Marwa 54, Sara 33, Nadia 37, Saba 47, Amna 28. Find the average runs of the students.

Sum = 21 + 52 + 54 + 33 + 37 + 47 + 28 = 272.

Number of students = 7.

Average = 272 / 7 = 38.86 runs (approximately).

## Exercise 2 - Bar Graphs (book pages 162-168)

Drawing a bar graph: The table shows the marks obtained out of 10 marks in the Mathematics test during the last 4 months: September 8, October 6, November 10, December 9. What is the procedure to draw the bar graph?

Mark the horizontal line as X-axis and the vertical line as Y-axis.

Write the names of the months on the X-axis and the marks obtained on the Y-axis.

One square on the Y-axis represents 1 mark.

8 marks were obtained in September, so colour 8 squares along the Y-axis.

Similarly, 6 marks in October, 10 marks in November and 9 marks in December.

In a bar graph, the width of each bar is the same. The bar graph given above is a vertical bar graph. Similarly, we can draw a horizontal bar graph.

Horizontal bar graph: The table shows the number of cars of different colours in a car parking during a month: Blue 80, Black 65, Grey 90, Red 75. Represent this data in a horizontal bar graph.

Mark the horizontal line as X-axis and the vertical line as Y-axis.

Write the number of cars on the X-axis and the colours of cars on the Y-axis.

One square on the X-axis represents 10 cars.

Draw the bars: Blue 80 (8 squares), Black 65 (6.5 squares), Grey 90 (9 squares), Red 75 (7.5 squares).

Reading a bar graph: The bar graph shows the number of flowers that bloomed in Huma's garden during five weeks. Read the graph.

6 flowers bloomed during the first week.

4 flowers bloomed during the second week.

3 flowers bloomed during the third week.

2 flowers bloomed during the fourth week.

7 flowers bloomed during the fifth week.

The maximum flowers bloomed during the fifth week.

The minimum flowers bloomed during the fourth week.

Reading a bar graph: The bar graph shows shoe production at a factory. In which year did the factory produce the most pairs of shoes? In which year the least? In which years did the factory make 20,000 pairs of shoes? How many less pairs of shoes were produced in 2011 than in 2015? In which 2 years is the difference in the number of shoes 10,000? How many pairs of shoes did the factory produce in these 5 years?

The factory produced the most pairs of shoes in 2015 (25,000 pairs).

The factory produced the least pairs of shoes in 2013 (15,000 pairs).

The factory made 20,000 pairs of shoes in 2011 and 2012.

25,000 - 20,000 = 5,000 fewer pairs of shoes were produced in 2011 than in 2015.

The difference is 10,000 between 2013 (15,000) and 2015 (25,000).

Total in 5 years = 20,000 + 20,000 + 15,000 + 22,500 + 25,000 = 102,500 pairs of shoes.

Q1. Saba asked about the most favourite fruit of each student in her class. The table is: Strawberry 5, Apple 4, Banana 8, Mango 9. Draw a horizontal bar graph of this data.

Q2. The statistics of patients visiting a hospital during 5 months is given below: February 2, March 6, April 10, May 12, June 15. Draw a vertical bar graph of the given data.

Q3. During a visit to a zoo the children saw different numbers of animals: Lion 4, Elephant 2, Monkey 8, Cheetah 3, Snake 12, Giraffe 6, Zebra 10. Draw its vertical bar graph.

Q4. In the table, the values of temperature in Lahore during a week of August are given: Monday 32 degrees C, Tuesday 33 degrees C, Wednesday 30 degrees C, Thursday 32 degrees C, Friday 36 degrees C, Saturday 40 degrees C, Sunday 38 degrees C. Draw a horizontal bar graph.

Q5. Study the graph (Transportation used for going to school: Bus 100, Rickshaw 60, Car 70, Motorbike 20, Walk 80) and answer: (a) How many students come to school by car? (b) Is the number of students coming to school on motorbikes less than those coming by rickshaw? If yes, then how much less? (c) Which means of transport do the most students use and what is the number of students coming to school by this means of transport? (d) What is the total number of students coming by car, rickshaw and bus altogether? (e) Tell the number of students coming to school on foot.

(a) 70 students come to school by car.

(b) Yes, motorbike (20) is less than rickshaw (60) by 60 - 20 = 40 students.

(c) The most students use the bus: 100 students.

(d) Car + Rickshaw + Bus = 70 + 60 + 100 = 230 students.

(e) 80 students come to school on foot (Walk).

Q6. Study the graph (Pet animals: Cat 9, Goat 4, Cow 8, Buffalo 7) and answer: (a) How many students have cat as a pet? (b) How many students have goat as a pet? (c) Which is the most popular pet? (d) In total, how many students have a pet at home? (e) If the total number of students in the class is 35, find out how many students do not have any pets in their home?

(a) 9 students have cat as a pet.

(b) 4 students have goat as a pet.

(c) Cat is the most popular pet (9 students).

(d) Total students with a pet = 9 + 4 + 8 + 7 = 28.

(e) Students without pets = 35 - 28 = 7.

## Review Exercise - Unit 9 (book pages 169-170)

Q1. (a) The average of a number of items can be found by:

✓ (i) Dividing sum of items by number of items

(ii) Adding sum of items and number of items

(iii) Multiplying sum of items by number of items

(iv) Subtracting sum of items from number of items

(b) If a student got 19, 21, 22, 24 and 19 marks in different subjects in the monthly test, his average marks will be:

(i) 19

✓ (ii) 21

(iii) 22

(iv) 25

(c) To find the sum of the given items whose average is known, the formula used is:

(i) Sum of items = average of items + number of items

(ii) Sum of items = average of items - number of items

✓ (iii) Sum of items = average of items x number of items

(iv) Sum of items = average of items / number of items

(d) If the sum of some quantities is 600 and the average is 50, then number of quantities will be:

(i) 15

✓ (ii) 12

(iii) 10

(iv) 5

(e) Ahmed jumped 12 times in a minute, 9 times in the second minute and 15 times in the third minute. What will be the average of the number of jumps Ahmed did?

(i) 9

(ii) 11

✓ (iii) 12

(iv) 15

Q2. A labourer earned Rs. 1,200 on the first day, Rs. 1,000 on the second day, Rs. 1,500 on the third day, Rs. 1,300 on the fourth day and Rs. 1,200 on the fifth day. Find out how many rupees he earned on average in five days?

Sum = 1,200 + 1,000 + 1,500 + 1,300 + 1,200 = 6,200.

Number of days = 5.

Average = 6,200 / 5 = Rs. 1,240 per day.

Q3. Find out the average of six odd numbers from 1 to 11 and also find the average of six even numbers from 2 to 12. Tell which of these two averages is greater in value, the even one or the odd one?

Six odd numbers from 1 to 11: 1, 3, 5, 7, 9, 11. Sum = 36. Average = 36 / 6 = 6.

Six even numbers from 2 to 12: 2, 4, 6, 8, 10, 12. Sum = 42. Average = 42 / 6 = 7.

The even average (7) is greater than the odd average (6).

Q4. A pharmacy earned a profit of Rs. 50,000 in the first month, Rs. 62,000 in the second month, Rs. 68,000 in the third month, Rs. 78,000 in the fourth month and Rs. 65,000 in the fifth month. Draw a vertical bar graph of the profit of the company for the five months.

Q5. The students of class-5 were asked to vote for their most favourite subjects. The bar graph shows: Urdu 5, English 6, Arts 10, Science 6, Mathematics 8, Social Studies 3. Answer: (a) How many students like Science and Mathematics? (b) How many students like Arts? (c) How many more students like Mathematics than Science as their favourite subject? (d) Which subjects are liked by the same number of students and what is their number? (e) Study the graph to find out the total number of students in class-5. (f) Tell the most favourite and the least favourite subject of class-5 students.

(a) Science + Mathematics = 6 + 8 = 14 students.

(b) 10 students like Arts.

(c) Mathematics - Science = 8 - 6 = 2 more students like Mathematics than Science.

(d) English and Science are liked by the same number of students: 6 students each.

(e) Total students = 5 + 6 + 10 + 6 + 8 + 3 = 38.

(f) Most favourite subject: Arts (10 students). Least favourite subject: Social Studies (3 students).